STABILITY TESTING OF DRUG SUBSTANCE

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DECLARATION

I hereby declare that the dissertation entitled **"Stability Testing of Drug Substance"** *is a bonafide record of the project work carried out by me under the supervision of* ***Dr. P. SANJEEVA.*** *No part of this dissertation has been presented earlier for the award of any other degree or diploma of this or any other university.*

Place: Ananthapuramu **(DUDEKULA MAHAMMAD RAFI)**

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CERTIFICATE

This is to certify that the dissertation entitled **"Stability Testing of Drug Substance"** by **DUDEKULA MAHAMMAD RAFI** for the award of the degree of Master of Science in chemistry is based on the results of experiments carried out by the candidate in the Department of Chemistry. No part of the dissertation has been previously submitted for any other degree or diploma of this or any other university.

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1.

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Examiners

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1. Introduction

Every drug product on the market has to apply for registration. One of the most important parts of the several hundred page documentation of the registration process is the detailed description and the evaluation of the stability of the pharmaceutical product. It is very important for the pharmaceutical companies to establish accurate, reliable and the longest possible shelf-life for their drug products. The adequate and properly testing methods help to reach these requirements.

1.1. Importance of Stability:

Although drug discovery gets much of the attention in the pharmaceutical industry, stability plays an equally important fundamental role. A drug must remain potent and safe long after it leaves a manufacturing facility. Ensuring this safety requires a range of technologies and applications.

Stability is defined as the capacity of a drug substance or a drug product to remain within specifications established to ensure its identity, strength, quality, and purity throughout the re-test period or expiration dating period, as appropriate in ICH Guidelines. The expiration dating period (shelf-life) is defined as the time interval until the drug product is expected to remain within the specifications established for its strength, quality and purity.

The purpose of the stability study is to provide evidence on the variation of quality of a drug product with time under the influence of a variety of environmental factors such as temperature, humidity and light, and to establish the expiration dating period (shelf life) for the drug product and for the recommended storage conditions. Overall, stability tests must confirm that a drug remains active—meaning that its structure doesn’t change over time and it must not create toxic degradation products.

1.2. Stability testing of API

1.2.1 Active Pharmaceutical Ingredient (API):

- The active ingredient, also known as the Active Pharmaceutical ingredient (API), is the substance in a drug that is biologically active. Some medications may contain more than one active ingredient. The traditional word for the API is ‘pharm Akon’ which originally denoted a magical substance or drug. A dosage form of the drug is traditionally composed of two things: The API and the excipient. The API is suspended in another material that is pharmaceutically inert (excipient). Impurity

profile is a description of the identified and unidentified impurities present in a typical batch of API (Active Pharmaceutical Ingredient) produced by a specific controlled production process. For each API there should be an impurity profile describing the identified and unidentified impurities present in a typical batch. The impurity profile is normally dependent upon the process or origin of the API.

- A number of recent articles have described a designed approach and guidance for isolating and identifying process-related impurities and degradation products using Mass Spectrometry (MS), Nuclear Magnetic Resonance (NMR), and High Performance Liquid Chromatography (HPLC) for pharmaceutical substances. In general, according to ICH (International Conference on Harmonization of Technical Requirements for Registration of Pharmaceuticals for Human Use) guidelines on impurities in new drug products, identification of impurities below.
- The 0.1% level is not considered to be necessary unless the potential impurities are expected to be unusually potent or toxic. In all cases, impurities should be qualified. According to ICH, the threshold for the various impurities is based on their daily dosage and is as follows,
- A key component of the overall quality of a pharmaceutical product is control of impurities, as even in small amounts they may affect drug safety and efficiency.
- The identification and quantification of impurities to acceptable standards presents a significant challenge to the analytical chemist. Analytical science is developing rapidly and provides increasing opportunity to identify the structure of these impurities.
- Qualification of impurity is the process of acquiring and evaluating data that establishes the biological safety of an individual impurity or a given impurity profile at the level(s) being considered. When appropriate, it is recommended that drug manufacturers provide a rationale for establishing impurity acceptance criteria that includes safety considerations.
- An impurity is considered qualified when it meets one or more of the following conditions:
- When the observed level and proposed acceptance criteria for the impurity do not exceed the level observed in an FDA-approved human drug product.
- When the impurity is a significant metabolite of the drug substance.

Classification of API

API can be broadly classified as:

- Inorganic substances;
- Organic substances (isolated from materials of animal or human origin)
- Organic substances (synthetic or semi-synthetic or isolated from herbal sources or micro-organisms).

Because of the concerns for safety, efficacy and quality, the regulatory agencies have found it appropriate to require stability testing for API. More recently, the process of globalization and harmonization has stimulated the development of worldwide standards. The International Conference on Harmonization (ICH) has set up draft guidelines for the "Stability testing of drug substances and drug products".

1.3 Relevant guidelines

Many countries have their own guidelines concerning stability testing. If a manufacturer wishes to market a product in several countries, it is simpler to use one of the international guidelines, such as those of WHO & ICH. Most countries will accept data generated according to ICH guidelines. Many countries will accept data generated according to WHO guidelines, & especially when the product in question has been prequalified by WHO - But possibly not ICH countries. Whilst ICH guidelines are more detailed than those of WHO, there are few 'in-principle' differences, except in relation to testing conditions for hot & humid climates.

1.4 Available guidelines

ICH Q1A (R2)-Stability testing of new drug substances and products

ICH Q1B-Photostability testing of new drug substances and products

ICH Q1C-Stability testing for new dosage forms

ICH Q1D-Bracketing and matrixing designs for stability testing of new drug substances and drug product

ICH Q1E-Evaluation of stability data

Major focus of this project work is to investigate stability of drug substances (API) as per ICH Q1A (R2).

1.4.1 General Principles of ICH Q1 (R2)

The purpose of stability testing is to provide evidence on how the quality of a drug substance or drug product varies with time under the influence of a variety of environmental

factors, such as temperature, humidity, and light, and to establish a retest period for the drug substance or a shelf life for the drug product and recommended storage conditions.

The choice of test conditions defined in this guidance is based on an analysis of the effects of climatic conditions in the three regions of the EU, Japan, and the United States. The mean kinetic temperature in any part of the world can be derived from climatic data, and the world can be divided into four climatic zones, I-IV.

Table 1: Different Climatic zones

Zone	Climatic zone	Mean Kinetic Temperature
Zone-I	**Temperate**	**21°C/45% RH**
Zone-II	**Mediterranean, subtropical**	**25°C/60% RH**
Zone-III	**Hot and dry**	**30°C/35% RH**
Zone-IVA	**Hot and humid**	**30°C/65% RH**
Zone-IVB	**Hot and very humid**	**30°C/75% RH**

This guidance addresses climatic zones I and II. The principle has been established that stability information generated in any one of the three regions of the EU, Japan, and the United States would be mutually acceptable to the other two regions, provided the information is consistent with this guidance and the labelling is in accord with national/regional requirements.

1.4.2 CH Q1A (R2)-Guidelines for Drug Substance:

General

Information on the stability of the API is an integral part of the systematic approach to stability evaluation.

Stress Testing: Stress testing of the drug substance can help identify the likely degradation products, which can in turn help establish the degradation pathways and the intrinsic stability of the molecule and validate the stability indicating power of the analytical procedures used.

The nature of the stress testing will depend on the individual drug substance and the type of drug product involved. Stress testing is likely to be carried out on a single batch of the drug substance. The testing should include the effect of temperatures (in 10°C increments (e.g., 50°C, 60°C) above that for accelerated testing), humidity (e.g., 75 percent relative humidity or great where appropriate, oxidation, and photolysis on the drug substance. The testing should also evaluate the susceptibility of the drug substance to hydrolysis across a wide range of pH values when in solution. Photo stability testing should be an integral part of stress

testing. The standard conditions for photo stability testing are described in ICH Q1B Photo stability Testing of New Drug Substances and Products.

Examining degradation products under stress conditions is useful in establishing degradation pathways and developing and validating suitable analytical procedures. However, such examination may not be necessary for certain degradation products if it has been demonstrated that they are not formed under accelerated or long-term storage conditions.

Results from these studies will form an integral part of the information provided to regulatory authorities.

Selection of Batches

Data from formal stability studies should be provided on at least three primary batches of the drug substance. The batches should be manufactured to a minimum of pilot scale by the same synthetic route as production batches and using a method of manufacture and procedure that simulates the final process to be used for production batches. The overall quality of the batches of drug substance placed on formal stability studies should be representative of the quality of the material to be made on a production scale.

Container Closure System

The stability studies should be conducted on the drug substance packaged in a container closure system that is the same as or simulates the packaging proposed for storage and distribution.

Specification

Specification, which is a list of tests, references to analytical procedures, and proposed acceptance criteria, is addressed in ICH Q6A and Q6B. In addition, specification for degradation products in a drug substance is discussed in ICH Q3A. Stability studies should include testing of those attributes of the drug substance that are susceptible to change during storage and are likely to influence quality, safety, and/or efficacy. Following table shows the changes in drug substances which are observed and measured by temperature, light and humidity.

Table 2: Changes observed and measured in drug substances

Physical changes	Appearance
	Melting point
	Clarity and colour of solution
	Water content
	Water content
	Crystal modification (polymorphism)
	Particle size
Chemical changes	Increase in degradation products
	Decrease of assay
Microbial changes	Growth of micro organism

The testing should cover, as appropriate, the physical, chemical, biological, and microbiological attributes.

Validated stability-indicating analytical procedures should be applied. Whether and to what extent replication should be performed should depend on the results from validation studies.

Testing frequency:

Long-term: Every 3 month 1st year, every 6 month second year and annually, thereafter ***Accelerated:*** A minimum of 3 time points including initial and final (0, 3, and 6 month) ***Intermediate storage*** if needed: minimum 4 time points including initial and final (0, 6, 9 and 12 months)

Storage conditions

In general an API should be evaluated under storage conditions (with appropriate tolerances) that test its thermal stability and, if applicable, its sensitivity to moisture. The storage conditions and the lengths of studies chosen should be sufficient to cover storage and shipment. Storage condition tolerances are defined as the acceptable variations in temperature

and relative humidity of storage facilities for stability studies. The equipment used should be capable of controlling the storage conditions within the ranges defined in these guidelines. The storage conditions should be monitored and recorded. Short-term environmental changes due to opening the doors of the storage facility are accepted as unavoidable. The effect of excursions due to equipment failure should be assessed, addressed and reported if judged to affect stability results. Excursions that exceed the defined tolerances for more than 24 hours should be described in the study report and their effects assessed. The long-term testing should normally take place over a minimum of 12 months for the number of batches at the time of submission, and should be continued for a period of time sufficient to cover the proposed re-test period or shelf-life. For existing substances that are known to be stable, data covering a minimum of six months may be submitted. Additional data accumulated during the assessment period of the registration application should be submitted to the authorities upon request. Data from the accelerated storage condition and, if appropriate, from the intermediate storage condition can be used to evaluate the effect of short term excursions outside the label storage conditions (such as might occur during shipping). Long-term, accelerated and, where appropriate, intermediate storage conditions for APIs are detailed below. The general case applies if the API is not specifically covered by a subsequent section. Alternative storage conditions can be used if justified. If long-term studies are conducted at 25 °C ± 2 °C/60% RH ± 5% RH and "significant change" occurs at any time during six months' testing at the accelerated storage condition, additional testing at the intermediate storage condition should be conducted and evaluated against significant change criteria. In this case, testing at the intermediate storage condition should include all long-term tests, unless otherwise justified, and the initial application should include a minimum of six months' data from a 12-month study at the intermediate storage condition.

Data on refrigerated storage should be assessed according to the evaluation section of these guidelines, except where explicitly noted below. If significant change occurs between three and six months' testing at the accelerated storage condition, the proposed re-test period should be based on the data available at the long-term storage condition.

If significant change occurs within the first three months' testing at the accelerated storage condition a discussion should be provided to address the effect of short-term excursions outside the label storage condition, e.g. during shipping or handling. This discussion can be supported, if appropriate, by further testing on a single batch of the API for a period shorter than three months but with more frequent testing than usual. It is considered unnecessary to

continue to test an API for the whole six months when a significant change has occurred within the first three months.

In the rare case of any API of non-biological origin being intended for storage in a freezer, the re-test period or shelf-life should be based on the long-term data obtained at the long-term storage condition. In the absence of an accelerated storage condition for APIs intended to be stored in a freezer, testing on a single batch at an elevated temperature (e.g. 5 °C ± 3 °C or 25 °C± 2 °C or 30 °C ± 2 °C) for an appropriate time period should be conducted to address the effect of short-term excursions outside the proposed label storage condition, e.g. during shipping or handling.

"Significant change" for an API is defined as failure to meet its specification.

Table 3: Storage conditions for stability of drug substances

Proposed Storage	Study Type	Storage Condition	Time (months)
Controlled room temp.	Long term	25°C ± 2°C / 60% RH ±5% RH	**12**
	Intermediate	30°C ± 2°C / 65% RH ±5°C	**6**
	Accelerated	40° C ± 2°C / 75% RH ± 5 % RH	**6**
Refrigerator	Long term	5° C ± 3ºC	**12**
	Accelerated	25°C ± 2ºC / 60% RH ± 5 %RH	**6**
Freezer	Long term	-20° C ± 5 °C	**12**

Stability Commitment

When available long-term stability data on primary batches do not cover the proposed retest period granted at the time of approval, a commitment should be made to continue the stability studies post approval to firmly establish the retest period. Where the submission includes long-term stability data on three production batches covering the proposed retest period, a post approval commitment is considered unnecessary. Otherwise, one of the following commitments should be made:

- If the submission includes data from stability studies on at least three production batches, a commitment should be made to continue these studies through the proposed retest period.

- If the submission includes data from stability studies on fewer than three production batches, a commitment should be made to continue these studies through the proposed retest period and to place additional production batches, to a total of at least three, on long-term stability studies through the proposed retest period.
- If the submission does not include stability data on production batches, a commitment should be made to place the first three production batches on long term stability studies through the proposed retest period. The stability protocol used for long-term studies for the stability commitment should be the same as that for the primary batches, unless otherwise scientifically justified.

Evaluation

The purpose of the stability study is to establish, based on testing a minimum of three batches of the drug substance and evaluating the stability information (including, as appropriate, results of the physical, chemical, biological, and microbiological tests), a retest period applicable to all future batches of the drug substance manufactured under similar circumstances. The degree of variability of individual batches affects the confidence that a future production batch will remain within specification throughout the assigned retest period. The data may show so little degradation and so little variability that it is apparent from looking at the data that the requested retest period will be granted. Under these circumstances, it is normally unnecessary to go through the formal statistical analysis; providing a justification for the omission should be sufficient.

An approach for analysing the data on a quantitative attribute that is expected to change with time is to determine the time at which the 95 percent, one-sided confidence limit for the mean curve intersects the acceptance criterion. If analysis shows that the batch-tobatch variability is small, it is advantageous to combine the data into one overall estimate. This can be done by first applying appropriate statistical tests (e.g., p values for level of significance of rejection of more than 0.25) to the slopes of the regression lines and zero time intercepts for the individual batches. If it is inappropriate to combine data from several batches, the overall retest period should be criteria. The nature of any degradation relationship will determine whether the data should be transformed for linear regression analysis. Usually the relationship can be represented by a linear, quadratic, or cubic function on an arithmetic or logarithmic scale. Statistical methods should be employed to test the goodness of fit of the data on all batches and combined batches (where appropriate) to the assumed degradation line or curve. Limited extrapolation of the real time data from the long-term storage condition

beyond the observed range to extend the retest period can be undertaken at approval time if justified. This justification should be based, for example, on what is known about the mechanism of degradation, the results of testing under accelerated conditions, the goodness of fit of any mathematical model, batch size, and/or existence of supporting stability data. However, this extrapolation assumes that the same degradation relationship will continue to apply beyond the observed data. Any evaluation should cover not only the assay, but also the levels of degradation products and other appropriate attributes.

Statements/Labelling

A storage statement should be established for the labelling in accordance with relevant National/regional requirements. The statement should be based on the stability evaluation of the drug substance. Wherever applicable, specific instructions should be provided, particularly for drug substances that cannot tolerate freezing. Terms such as ambient conditions or room temperature should be avoided. A retest period should be derived from the stability information, and a retest date should be displayed on the container label if appropriate.

1.5. Analytical Techniques and Instrument used in Industry:

Modern analytical chemistry is dominated by instrumental analysis. There are so many different types of instruments today that it can seem like a confusing array of acronyms rather than a unified field of study. Many analytical chemists focus on a single type of instrument. Academics tend to either focus on new application and discoveries or on new methods of analysis. It plays an increasing important role in the pharmaceutical industry where it is used in the discovery of new drug candidates and in development applications.

There are two branches in analytical chemistry:

1. **Qualitative analysis:** Qualitative analysis is the determination of those elements and compounds that are present in a sample of unknown materials.
2. **Quantitative analysis:** Quantitative analysis is the determination of the amount by weight of each element or compound present.

Classification of Analytical Techniques:

Analytical methods are often being classified as either:

(1) Traditional analytical technique

(2) Instrumental analysis

Traditional Analytical technique:

Although modern analytical chemistry is dominated by sophisticate instrumentation, the roots of analytical chemistry and some of the principles used these techniques also tend to form the backbone of the most undergraduate analytical chemistry educational labs. Example includes technique like titration, gravimetric analysis, quantitative analysis, etc.

Instrumental Analysis:

Most of the major developments in analytical chemistry take place after 1900. During this period instrumental analysis becomes progressively dominant in the field. In particular many of the basic spectroscopic and spectrometric techniques were discovered in the early 20th century and refined in the late 20th century. The separation sciences follow a similar time line of development and also become increasing transformed into high performance instruments. In the 90th many of these techniques began to be used together to achieve a complete characterization of sample. Example includes technique like

- Spectroscopy
- Mass Spectroscopy
- Crystallography
- Electrochemical analysis
- Thermal analysis
- Separation
- Hybrid Technique

Introduction of Analytical chemistry:

Analytical chemistry is a measurement science consisting of a set of powerful ideas and methods that are useful in all fields of sciences and medicine. Analytical chemistry mainly deals with the qualitative and quantitative determination of substance. For that Analytical methods are used. On the basis of substance property these methods are:-

ANALYTICAL METHODS:

- **PHYSICAL**
- **CHEMICAL** ⟹ **1. Qualitative**
  2. Quantitative
  3. Structural

Chemical analysis has some basic steps:

1. Choice of Method

2. Sampling

3. Primary sample treatment

4. Separation

5. Final Measurement

6. Presentation of data

CHROMATOGRAPHY ⟹ 1. Planar chromatography

(a) Gas Chromatography

(b) Thin Layer Chromatography

(c) High Performance Liquid Chromatography

(d) Liquid Chromatography

2. Column chromatography

Introduction of Liquid Chromatography (LC):

Liquid chromatography (LC) is a physical separation technique conducted in the liquid phase. A sample is separated into its constituent components (or analytes) by distributing between the mobile phase (a flowing liquid) and a stationary phase (sorbents packed inside a column). For example, the flowing liquid can be an organic solvent such as Acetonitrile, Methanol and the stationary phase can be porous silica particles packed in a column. HPLC is a modern form of LC that uses small-particle columns through which the mobile phase is pumped at high pressure is a schematic of the chromatographic process, where a mixture of analyte A and B separated into two distinct bands as they migrate down the column filled with packing (stationary phase) is a representation of the dynamic partitioning process of the analyte between the flowing liquid and a spherical packing particle. Note that the movement of component B is retarded in the column because each B molecule has stronger affinity for the stationary phase than the A molecule. An in-line detector monitors the concentration of each separated component band in the effluent and generates a trace called the chromatogram.

2. Instrumentation

Recent days the current guidelines and regulatory requirements are very stringent in terms of estimation and quantification of residual impurities in any of synthetic route or process, due to this reason the separation of all compounds in single run or during analysis, the LC methods was taken preference over regular conservative mode methods of analysis. The advantage of LC methods is that the developed analytical methods posses greater selectivity, sensitivity, accurate, precise and robust. In this category the various kinds of LC methods that have been used are TLC, HP-TLC, GC, HPLC and UPLC. In comparison, HPLC and UPLC recently have been generally working. Due to the specificity, sensitivity and greater resolution between all pairs of compounds therefore it has increased recognition in stability indicting methodologies. We can find the solution or analysing these type analysis i.e. non–volatiles, thermal sensitive, high polar nature of products by using above new and conventional techniques. Therefore, most of the methods were developed by using UPLC and HPLC. The control of drug substances impurities is currently a serious subject to the pharmaceutical industry. The bi-products in pharmaceuticals are the unwanted chemicals that leftovers at residual levels with the APIs, develop during the synthesis route and upon preparation of drug products. The attendance of these residual Impurities may influence the safety and efficacy of bulk products. The composition of impurities allows one to draw conclusions regarding the manufacturing of the drug products and its adulteration, which is becoming pervasive in all countries of the world therefore, it is strictly to firmly manage the safety of products and to determine the content of impurities at all stages of production from raw materials to finished medicinal forms.

2.1 Importance of impurity evaluation in pharmaceutical industry

Pharmaceutical industry is emergent day by day with the aim to develop new drugs extracted from synthetically or natural products produced chemical substances. But one thing always remains important that the product/substances should be the best pure as possible. Therefore, purity (assay) has been always considered as an essential factor to ensuring drug quality and its safety. To purify a substance and remove the excess impurities one should first identify that whether they are actually present and what their nature is.

In the past, this was not always done, because for non availability of advanced techniques. But presently drug analysis and pharmaceutical impurities are the subjects of

continuous review in the public concern. The ICH guidelines achieved an enormous deal in harmonizing the definitions of the impurities in drug substances.

2.2 The role of chromatography in impurity evaluation

The applications of chromatography have grown explosively in the last fifty years, owing not only to the development of several new types of chromatographic techniques introduced or available in the market, but also to the growing need by scientists for better methods for characterizing complex molecules. Indeed the greatest advantage of the chromatographic method over any other analytical procedure is the ability of separating specific analytes, a feature that appeals to all branches of science, which enables to discover and analyze unknown elements and chemical compounds.

2.3 Analysis through chromatography

The technique through which the chemical components present in complex mixtures are separated, identified and determined is termed as chromatography. This technique is widely used in most of fields like Pharmaceuticals, Biochemical, Food products, Industrial chemicals, Pollutants, Forensic chemistry and is a very powerful tool not only for analytical methods but also for preparative methods. Compounds of high grade purity can be obtained by this method. Chromatography can be simply defined as follows: “It is the technique in which the components of a mixture are separated based upon the rates at which they are carried or moved through a stationary phase (column) by a gaseous or liquid mobile phase”.
"Chromatography" represents a separation technique; whereas a “**Chromatograph"** is a system for performing chromatography. The chart displaying the time-dependent change in signal intensity as a result of the separation is called a **"chromatogram”**.

2.4 High performance Liquid chromatography (HPLC)

2.4.1 Principle of HPLC:

High-performance liquid chromatography (formerly referred to as high-pressure liquid chromatography), HPLC, is a technique in analytic chemistry used to separate the components in a mixture, to identify each component, and to quantify each component. HPLC relies on pumps to pass a pressurized liquid (mobile phase) and a sample mixture through a column filled with a sorbent, leading to the separation of the sample components.

The active component of the column, the sorbent, is typically a granular material made of solid particles (e.g. silica, polymers, etc.), 2–50 micrometers in size. The components of the sample mixture are separated from each other due to their different degrees of interaction with the sorbent particles. The pressurized liquid is typically a mixture of solvents (e.g. water, acetonitrile and/or methanol) and is referred to as a "mobile phase". Its composition and temperature play a major role in the separation process by influencing the interactions taking place between sample components and sorbent. These interactions are physical in nature, such as hydrophobic (dispersive), dipole–dipole and ionic, most often a combination thereof.

HPLC is distinguished from traditional ("low pressure") liquid chromatography because operational pressures are significantly higher (50–350 bar) in HPLC, while ordinary liquid chromatography typically relies on the force of gravity to pass the mobile phase through the column. Due to the small sample amount separated in analytical HPLC, typical column dimensions are 2.1–4.6 mm diameter, and 30–250 mm length. Also HPLC columns are made with smaller sorbent particles (2–50 micrometer in average particle size). This gives HPLC superior resolving power when separating mixtures, which is why it is a popular chromatographic technique.

2.4.2 Types of HPLC

- **Normal-phase chromatography**
- **Reverse-phase chromatography**
- **Ion-exchange chromatography**
- **Size exclusion chromatography**
- **Bio affinity chromatography**

2.4.3 Normal–phase chromatography

Normal-phase HPLC (NP-HPLC) separates analytes based on their affinity for a polar stationary surface such as silica; hence it is based on analyte ability to engage in polar interactions (such as hydrogen-bonding or dipole-dipole type of interactions) with the sorbent surface. NP-HPLC uses a non-polar mobile phase (e.g. Chloroform), and works effectively for separating analytes readily soluble in non-polar solvents. The analyte associates with and is retained by the polar stationary phase. Adsorption strengths increase with increased analyte polarity. The interaction strength depends not only on the functional groups present in the structure of the analyst molecule, but also on steric factors. The effect of steric hindrance on

interaction strength allows this method to resolve (separate) structural isomers. The use of more polar solvents in the mobile phase will decrease the retention time of analytes, whereas more hydrophobic solvents tend to induce slower elution (increased retention times). Very polar solvents such as traces of water in the mobile phase tend to adsorb to the solid surface of the stationary phase forming a stationary bound (water) layer which is considered to play an active role in retention. This behavior is somewhat peculiar to normal phase chromatograhy because it is governed almost exclusively by an adsorptive mechanism (i.e. analytes interact with a solid surface rather than with the solvated layer of a ligand attached to the sorbent). Adsorption chromatography is still widely used for structural isomer separations in both column and thin-layer chromatography formats on activated (dried) silica or alumina supports.

2.4.4 Reversed-phase chromatography (RPC)

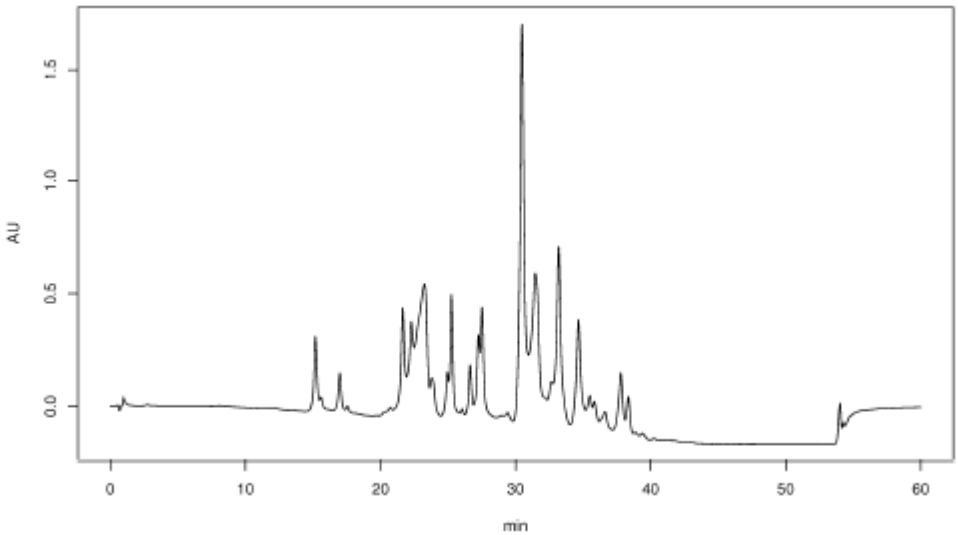


Figure 1: Chromatogram of complex mixture (perfume water) obtained by reversed phase HPLC

Reversed phase HPLC (RP-HPLC) has a non-polar stationary phase and polar mobile phase.

Stationary phase of RP-HPLC:

C-2 Ethyl Silyl – $Si\text{-}CH_2\text{-}CH_3$

C-8 Octyl Silyl – $Si\text{-}(CH_2)_7\text{-}CH_3$

C-18 Octadecyl Silyl – $Si\text{-}(CH_2)_{17}\text{–}CH_3$

CN Cyanopropyl Silyl – $Si\text{-}(CH_2)_3\text{-}CN$

Solvents (Mobile phase) used in RP-HPLC:

Methonal	CH_3OH
Acetonitrile	CH_3CN
Tetrahydrofuran	$(CH_2)_4O$
Water	H_2O

With such stationary phases, retention time is longer for molecules which are less polar, while polar molecules elute more readily (early in the analysis). An investigator can increase retention times by adding more water to the mobile phase; thereby making the affinity of the hydrophobic analyte for the hydrophobic stationary phase stronger relative to the now more hydrophilic mobile phase. Similarly, an investigator can decrease retention time by adding more organic solvent to the eluent. RP-HPLC is so commonly used that it is often incorrectly referred to as "HPLC" without further specification. The pharmaceutical industry regularly employs RP-HPLC to qualify drugs before their release. RP-HPLC operates on the principle of hydrophobic interactions, which originates from the high symmetry in the dipolar water structure and plays the most important role in all processes in life science. RP-HPLC allows the measurement of these interactive forces. The binding of the analyte to the stationary phase is proportional to the contact surface area around the non-polar segment of the analyte molecule upon association with the ligand on the stationary phase. This solvophobic effect is dominated by the force of water for "cavity-reduction" around the analyte and the C18-chain versus the complex of both. The energy released in this process is proportional to the surface tension of the eluent (water: 7.3×10^{-6} J/cm², methanol: 2.2×10^{-6} J/cm²) and to the hydrophobic surface of the analyte and the ligand respectively. The retention can be decreased by adding a less polar solvent (methanol, acetonitrile) into the mobile phase to reduce the surface tension of water. Gradient elution uses this effect by automatically reducing the polarity and the surface tension of the aqueous mobile phase during the course of the analysis. Structural properties of the analyte molecule play an important role in its retention characteristics. In general, an analyte with a larger hydrophobic surface area (C-H, C-C, and generally non-polar atomic bonds, such as S-S and others) is retained longer because it is non-interacting with the water structure. On the other hand, analytes with higher polar surface area (conferred by the presence of polar groups, such as -OH, $-NH_2$, COO^- or $-NH_3^+$ in their structure) are less retained as they are better integrated into water. Such interactions are subject to steric effects in that very large molecules may have only restricted access to the pores of the stationary phase, where the interactions with

surface ligands (alkyl chains) take place. Such surface hindrance typically results in less retention.

Retention time increases with hydrophobic (non-polar) surface area. Branched chain compounds elute more rapidly than their corresponding linear isomers because the overall surface area is decreased. Similarly organic compounds with single C-C-bonds elute later than those with a C=C or C-C-triple bond, as the double or triple bond is shorter than a single C-C-bond. Aside from mobile phase surface tension (organizational strength in eluent structure), other mobile phase modifiers can affect analyte retention. For example, the addition of inorganic salts causes a moderate linear increase in the surface tension of aqueous solutions (ca. 1.5×10^{-7} J/cm² per mol for NaCl, 2.5×10^{-7} J/cm² per mol for $(NH_4)_2SO_4$), and because the entropy of the analyte-solvent interface is controlled by surface tension, the addition of salts tend to increase the retention time. This technique is used for mild separation and recovery of proteins and protection of their biological activity in protein analysis (hydrophobic interaction chromatography, HIC). Another important factor is the mobile phase pH since it can change the hydrophobic character of the analyte. For this reason most methods use a buffering agent, such as sodium phosphate, to control the pH. Buffers serve multiple purposes: control of pH, neutralize the charge on the silica surface of the stationary phase and act as ion pairing agents to neutralize analyte charge. Ammonium formate is commonly added in mass spectrometry to improve detection of certain analytes by the formation of analyte ammonium adducts. A volatile organic acid such as acetic acid, or most Commonly formic acid is often added to the mobile phase if mass spectrometry is used to analyze the column effluent. Trifluoroacetic acid is used infrequently in mass spectrometry applications due to its persistence in the detector and solvent delivery system, but can be effective in improving retention of analytes such as carboxylic acids in applications utilizing other detectors, as it is a fairly strong organic acid. The effects of acids and buffers vary by application but generally improve chromatographic resolution. Reversed phase columns are quite difficult to damage compared with normal silica columns; however, many reversed phase columns consist of alkyl derivatized silica particles and should never be used with aqueous bases as these will destroy the underlying silica particle. They can be used with aqueous acid, but the column should not be exposed to the acid for too long, as it can corrode the metal parts of the HPLC equipment. RPHPLC columns should be flushed with clean solvent after use to remove residual acids or buffers, and stored in an appropriate composition of solvent. The metal content of HPLC columns must be kept low if the best possible ability to separate substances is to be retained. A good test for the metal content of a column is to

inject a sample which is a mixture of 2,2'- and 4,4'- bipyridine. Because the 2,2'-bipy can chelate the metal, the shape of the peak for the 2,2'-bipy will be distorted (tailed) when metal ions are present on the surface of the silica..

This technique is widely used for the molecular weight determination of polysaccharides.

SEC is the official technique (suggested by European pharmacopeia) for the molecular weight comparison of different commercially available low-molecular weight heparins.

2.4.5. Ion exchange chromatography:

In Ion- exchange chromatography, retention is based on the attraction between solute ions and charged sites bound to the stationary phase. Ions of the same charge are excluded. This form of chromatography is widely used in purifying water, Ligand- exchange chromatography, Ion-exchange chromatography of proteins, High-pH anion-exchange chromatography of carbohydrates and oligosaccharides etc.

2.4.6. Size exclusion chromatography:

Size exclusion chromatography (SEC), also called as gel permeation chromatography or gel filtration chromatography mainly separates particles on the basis of size. It is also useful for determining the tertiary structure and quaternary structure of proteins and amino acids. This technique is widely used for the molecular weight determination of polysaccharides.

2.4.7. Bio affinity chromatography:

Separation based on specific reversible interaction of proteins with ligands. Ligands are covalently attached to solid support on a bio-affinity matrix, retains proteins with interaction to the column-bound ligands. Proteins bound to a bio-affinity column can be eluted in two ways: Bio-specific elution: inclusion of free ligand in elution buffer which competes with column bound ligand. A-specific elution: change in pH, salt, etc. which weakens interaction protein with column-bound substrate. Because of specificity of the interaction, bio-affinity chromatography can result in very high purification in a single step.

2.5 Modes of HPLC:

3.5.1 Isocratic and gradient elution

A separation in which the mobile phase composition remains constant throughout the procedure is termed isocratic (meaning constant composition). The mobile phase composition does not have to remain constant. A separation in which the mobile phase composition is

changed during the separation process is described as a gradient elution. One example is a gradient starting at 10% methanol and ending at 90% methanol after 20 minutes. The two components of the mobile phase are typically termed "A" and "B"; A is the "weak" solvent which allows the solute to elute only slowly, while B is the "strong" solvent which rapidly elutes the solutes from the column. In reversed-phase chromatography, solvent A is often water or an aqueous buffer, while B is an organic solvent miscible with water, such as acetonitrile, methanol, THF, or isopropanol. In isocratic elution, peak width increases with retention time linearly according to the equation for N, the number of theoretical plates. This leads to the disadvantage that late eluting peaks get very flat and broad. Their shape and width may keep them from being recognized as peaks.

Gradient elution decreases the retention of the later-eluting components so that they elute faster, giving narrower (and taller) peaks for most components. This also improves the peak shape for tailed peaks, as the increasing concentration of the organic eluent pushes the tailing part of a peak forward. This also increases the peak height (the peak looks "sharper"), which is important in trace analysis. The gradient program may include sudden "step" increases in the percentage of the organic component, or different slopes at different times – all according to the desire for optimum separation in minimum time. In isocratic elution, the selectivity does not change if the column dimensions (length and inner diameter) change – that is, the peaks elute in the same order. In gradient elution, the elution order may change as the dimensions or flow rate change. The driving force in reversed phase chromatography originates in the high order of the water structure. The role of the organic component of the mobile phase is to reduce this high order and thus reduce the retarding strength of the aqueous component.

2.6 Components of HPLC system

The components of a basic high-performance liquid chromatography [HPLC] system are shown in the simple diagram in Figure 2. These are as follows:-

- Solvent reservoir and solvent system
- High pressure pump
- Injector
- Column
- Detector

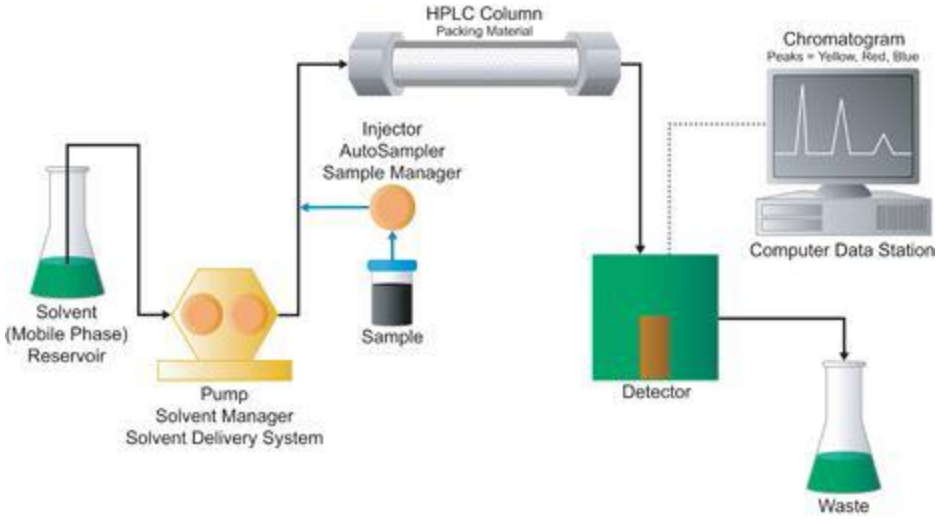


Figure 2: Schematic of HPLC System

2.6.1 Injector

Samples are injected into the HPLC/UPLC via an injection port. The injection port of an HPLC/UPLC commonly consists of an injection valve and the sample loop. The sample is typically dissolved in the mobile phase or in any of suitable solvents before injection into the sample loop. The sample solution is then drawn into a syringe and injected into the loop via the injection valve. A rotation of the valve rotor closes the valve and opens the loop in order to inject the sample solution into the stream of the mobile phase. Loop volumes can range between 200µl to over 1.0µl. In modern HPLC/UPLC systems, the sample injection is typically automated.

2.6.2 Column

Figure 3: Typical representation of column

LC columns are stainless steel tubes, typically of 5-30 cm in length and 1.7-5mm inner diameter. Guard columns, which are placed before an analytical column to trap junk and make longer the lifetime of the analytical columns, are 3-10 cm long. The stationary phase or packing is retained at each end by thin stainless steel frits with a mesh of 2μm or less. The most popular material is octadecyl-silica, which contains C18 chains, but materials with C2, C6, C8 and C22, chains are also available.

Types of columns in HPLC:

- Analytical (internal diameter (i.d) 1.0-4.6mm, length 15-250mm)
- Preparation (i.d > 4.6mm: lengths 50-250mm)
- Capillary (i.d. 0.1-1.0mm: various lengths)
- Nano (i.d < 0.1mm, or sometime stated as <100mm)

2.6.3 Detectors

Detector for HPLC/UPLC is the component that emits a response due to eluting sample compound and subsequently signals a peak on the chromatogram. It is positioned immediately posterior to the column stationary phase in order to detect the compounds as they elute from the column. The bandwidth and height of the peaks may usually be adjusted using the common and fine-tuning controls and the detection and sensitivity parameters may also be controlled. Regardless of the principle of operation, an ideal LC detector should have the following properties

(i) Low drift and noise level (particularly crucial in trace analysis)

(ii) High sensitivity

(iii) Fast response

(iv) Wide linear dynamic range (this simplifies quantization)

(v) Low dead volume (minimal peak broadening)

(vi) Cell design, which eliminates remixing of the separated bands

(vii)Insensitivity to changes in type of solvent, flow rate, and temperature

(viii) Operational simplicity and reliability

(ix) It should be tuneable so that detection can be optimized for different compounds and

(x) It should be non-destructive, many types of detectors can be used with HPLC/UPLC.

UV detectors:

These detector systems work on the spectroscopy principle. The sample detection depends on absorption of UV ray energy by the sample. The equipment comprises of accessories in order as UV source, grating (for light defraction),sample passing through a tubing exposed to rays, photo cell, charge conductor etc. When the UV rays emitted by lamp pass through gratings, rays split into different wavelengths. One specific wavelength rays are passed through sample. Some amount of light is absorbed by sample and the unabsorbed rays which fall on photo cell. These rays on collision on photo cell produce electrons whose current is recorded. This is indicative of nature and quantity of sample. This UV wavelength range of absorption is specific for sample.

They are capable to detect very wide range of compounds. The sensitivity ranges till microgram quantity of estimation. The schematic diagram of UV detector is shown in below figure:

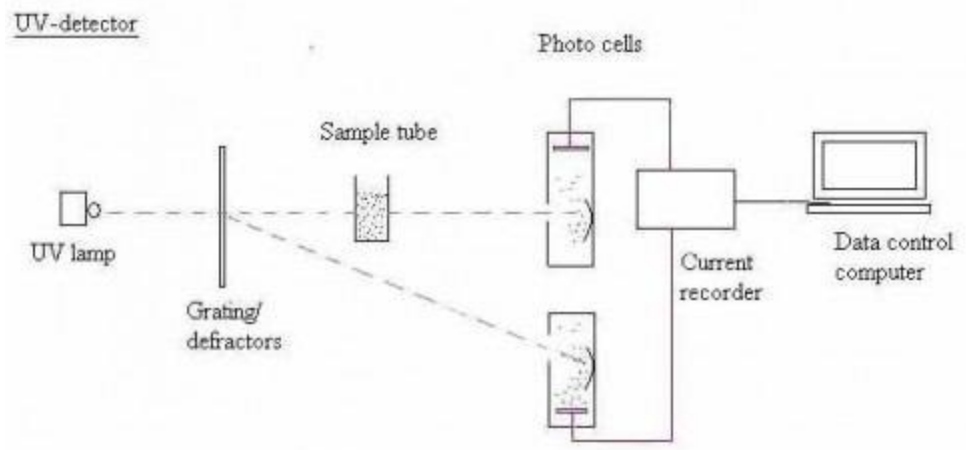


Figure 4: The schematic diagram of UV detector

Fixed wavelength: Measures at one wavelength, usually 254 nm.

Variable wavelength (Monochromator): Measures one wavelength (example: 210) at a time, but can detect over a wide range of wavelengths i.e 190 to 400 nm.

Multiple wavelengths (Photo Diode array): These are detectors which follow principle similar to UV detectors but the range of detection extends from UV, visible and to some extent to IR region. Thus the advantage is higher sensitivity and measures the entire absorption range i.e it gives scan of entire spectrum.

2.6.4 Fluorescent detectors

They measure the ability of a compound to absorb then re-emit light at given wavelengths. The excitation is set to the utmost value then the emission is scanned to locate the emission intensity. Selection of the initial system could, therefore, be based on assessment of the nature of test sample and drug substances.

2.6.5. Selection of Mobile Phase:

The mobile phase effects resolution, selectivity and efficiency. In Reverse Phase chromatography, the mobile phase consists of an aqueous buffer and a non-UV active water miscible organic solvent. The effect of the organic and aqueous phase and the proportions in which they are mixed will affect the analysis of the drug molecule. Typically a 10-50mM solution of an aqueous phase is H_3PO_4 in water i.e. phosphate buffer. The pH of a phosphate buffer is easily adjusted by using mono-di or tribasic phosphate salts. However, when phosphate salts are used the solution should be filtered to remove insoluble particles with 0.22µm filter paper. Other non-UV active acids and bases may also be used to effect differences in peak shape and retention 28.

2.6.6 Buffer Selection:

Choice of buffer is typically governed by the desired pH. The typical pH range for reversed phase on silica-based packing is pH 2 to 8. It is important that the buffer has a pKa close to the desired pH since buffer controls pH best at their pKa. A rule is to choose a buffer with a pKa value < 2 units of a desired mobile phase H.

2.7. Working of HPLC

A reservoir holds the solvent [called the mobile phase, because it moves]. A high-pressure pump [solvent delivery system or solvent manager] is used to generate and meter a specified flow rate of mobile phase, typically millilitres per minute. An injector [sample manager or auto sampler] is able to introduce [inject] the sample into the continuously flowing mobile phase stream that carries the sample into the HPLC column. The column contains the chromatographic packing material needed to effect the separation. This packing material is called the stationary phase because it is held in place by the column hardware. A detector is needed to *see* the separated compound bands as they elute from the HPLC column [most compounds have no color, so we cannot see them with our eyes]. The mobile phase

exits the detector and can be sent to waste, or collected, as desired. When the mobile phase contains a separated compound band, HPLC provides the ability to collect this fraction of the eluate containing that purified compound for further study. This is called preparative chromatography. Note that high-pressure tubing and fittings are used to interconnect the pump, injector, column, and detector components to form the conduit for the mobile phase, sample, and separated compound bands.

The detector is wired to the computer data station, the HPLC system component that records the electrical signal needed to generate the chromatogram on its display and to identify and quantitate the concentration of the sample constituents. Since sample compound characteristics can be very different, several types of detectors have been developed. For example, if a compound can absorb ultraviolet light, a UV-absorbance detector is used. If the compound fluoresces, a fluorescence detector is used. If the compound does not have either of these characteristics, a more universal type of detector is used, such as an evaporative-light-scattering detector [ELSD]. The most powerful approach is the use multiple detectors in series. For example, a UV and/or ELSD detector may be used in combination with a mass spectrometer [MS] to analyze the results of the chromatographic separation. This provides, from a single injection, more comprehensive information about an analyte. The practice of coupling a mass spectrometer to an HPLC system is called LC/MS.

Figure 5: Typical HPLC [Waters Alliance] System

2.8 Separation of components in column

The speed of a migrating sample component depends on whether the component has an affinity for the stationary or mobile phase. This affinity appears via various actions: adsorption, partition, ion exchange, etc. As shown in Figure 2, components that have a higher affinity for the mobile phase compared with the stationary phase migrate more rapidly, while components that have a higher affinity for the stationary phase are eluted from the column later. The order and resolution of the components emerging from the column depend on the type of selected stationary and mobile phases.

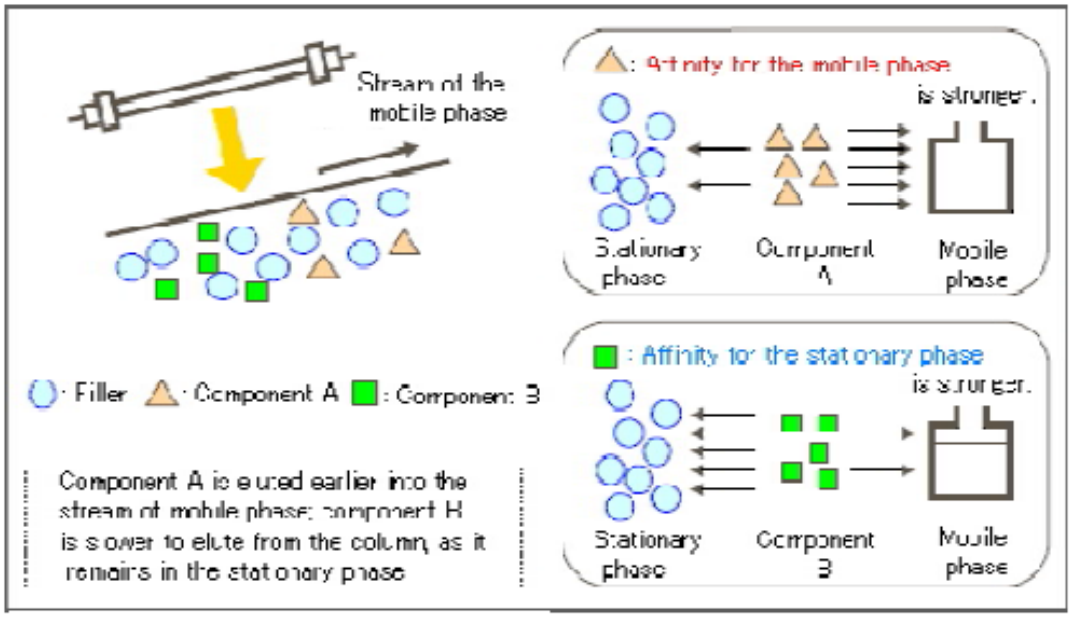


Figure 6: Diagram showing separation

2.8.1 Chromatogram of HPLC

Chromatogram is the output of HPLC. If we inject the sample mixture containing multiple components in HPLC, then multiple peaks come out as output signals as shown in Figure: 5.

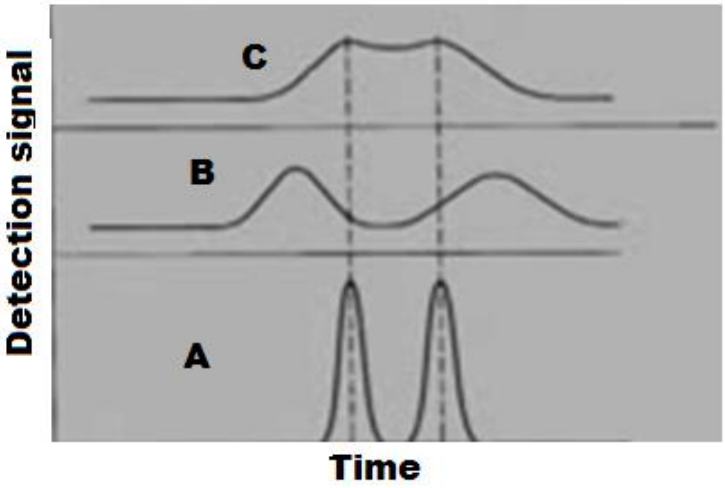


Figure 7: Chromatogram showing Optimizing separation

The Quality of separation is identified by sharpness of the peaks achieved during the analysis of sample by HPLC. Peaks coming out like peaks in region-A are considered excellent. Peaks in region-B are overlapping so it is really tough to differentiate areas of both peaks and while it is tougher to differentiate from peaks of region-C as they are overlapping each other. But region-A give the best separation, one can hope for this type of separation. This is very good base line separation. These two components are well separated, so that area of this curve can be measured independently without any error, and then concentration of these species. It's a challenge for an analyst to separate each component and clearly at base-line, and measures the area of the curve and hence determines the composition of that particular compound of his interest.

2.8.2 Retention time:

The time taken for a particular compound to travel through the column to the detector is known as its retention time. This time is measured from the time at which the sample is injected to the point at which the display shows a maximum peak height for that compound. Different compounds have different retention times. For a particular compound, the retention time will vary depending on:

- The pressure used (because that affects the flow rate of the solvent).
- The nature of the stationary phase (not only what material it is made of, but also particle size).
- The exact composition of the solvent.
- The temperature of the column.

2.9. Factor effecting on chromatography:

Column Dimension:

• Short (30-50mm) - short run times, low backpressure

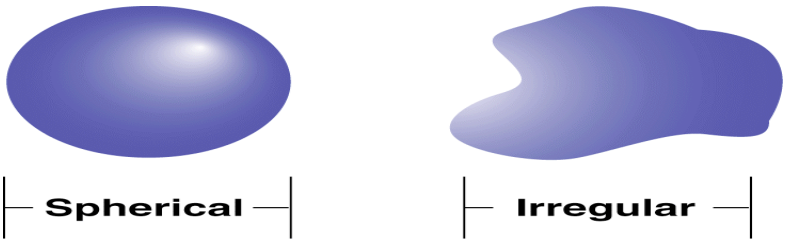
• Long (250-300mm) - higher resolution, long run times

• Narrow (≤ 2.1mm) - higher detector sensitivity

• Wide (10-22mm) - high sample loading

Particle shape:

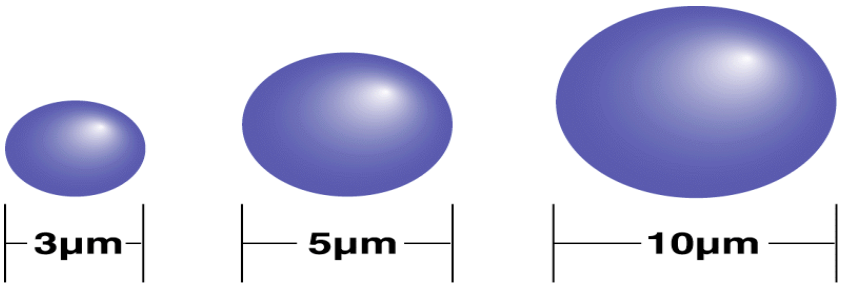
Spherical particles offer reduced back pressures and longer column life when using viscous mobile phases like 50:50 MeOH: H_2O.



Particle shape of column

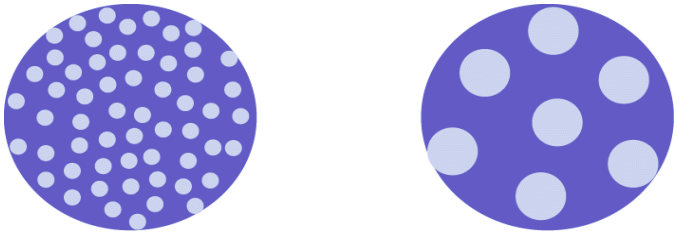
Particle size:

Smaller particles offer higher efficiency, but also cause higher backpressure. Choose 3µm particles for resolving complex, multi-component samples. Otherwise, choose 5 or 10µm packing.



Particle size of HPLC column

Surface area: High surface area generally provides greater retention, capacity and resolution for separating complex, multi-component samples. Low surface area packings generally equilibrate quickly, especially important in gradient analyses.



Surface area of column particle

HPLC Solvents:

The following table a series of commonly used HPLC solvents and their most partinent physical properties

Solvent	Dielectric Constant (25°C)	Polarity	UV cut off
Water	71	↑	190
Methanol	38		205
Acetonitrile	37.5		190
Dimethyl Formamide	37		268
Ethanol	25		210
Dichloromethane	9.1		220
Tetrahydrofuran	7.6		210
Hexane	1.9		200

HPLC Detectors:

In HPLC, the detector is selected in accordance with the analyte. Many of the detectors used are based on optical absorbance, such as UV-Visible absorbance detectors and photodiode array (PDA) detectors.

Absorbance detectors have the following features:

- The detector sensitivity and selectivity can be adjusted by setting on appropriate detector wavelength.
- The sensitivity is relatively high (although) it depends on the molar extinction coefficient.
- The stability is high.

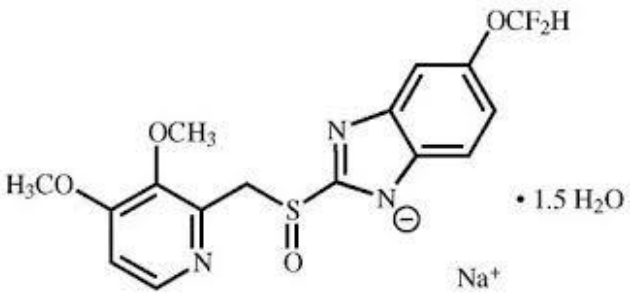
2.10. System Operation of HPLC:

- Preparation of mobile phase: Prepare the mobile phase as specified in STP.
- Treatment of Mobile phase: Reservoir of HPLC is equipped with a mean of removing dissolved gases.
- Sonicate
- Connect the mobile phase to the HPLC.
- Open the drain valve and purge it for 5 minutes or until all the air bubbles are removed. Because these air bubbles can cause band spreading & interference with detector.
- Switch on the pump and close the drain valve. Now the mobile phase runs through the column and whatever impurities are present in the column they are removed off. This is called the conditioning of the column.
- Switch on the computer.
- Select the method and parameters of the method.
- After getting a straight baseline, the mobile phase is injected in to the column to check any peak.
- Inject standard, Save and run the standard 3 times.
- Inject sample. Save and run.
- After getting peak we process them, and then take their printout

3. Case study

As a part of project work, Stability testing of selected drug substance has been carried out under long term storage conditions of temperature and humidity as per ICH guidelines. Drug substance has been subjected to various conditions as per mentioned in ICH guidelines for 3 months due to time constraints. Different parameters are monitored under influence of which drug substance is affected by humidity and temperature.

Product Name: X

CATEGORY	API
STRUCTURE	
CHEMICAL/IUPAC NAME	Sodium 5-(difluoromethoxy)-2-[(RS)-(3,4-dimethoxypyridin-2-yl)methyl]benzimidazol-1-ide sesquihydrate
MOLECULAR FORMULA	$C_{16}H_{14}F_2N_3NaO_4S.1.5H_2O$
MOLECULAR WEIGHT	432.4 g/mol
CAS NUMBER	164579-32-2
THERAPEUTIC CATEGORY	Anti-ulceractive, Proton pump inhibitor

3.0. Review of Literature:

- **Ali T. et al.,(2009)** reviewed regarding the concerns of long-term PPI use, a literature search was performed to identify pertinent original and review articles. Despite study shortcomings, the collective body of information overwhelmingly suggests an increased risk of infectious complications and nutritional deficiencies. Data regarding any increased risk in gastric or colon malignancy are less convincing. PPIs have revolutionized the management and complications of acid-related disorders with a high margin of safety; however, with the data available, efforts to reduce the dosing of or discontinue the use of PPIs must be reassessed frequently.
- **Dennis L.Avner (2000)** reviewed the pharmacological properties of this drug and summarized the findings from clinical studies of this drug. Due to its unique pharmacokinetic properties, mechanism of action, and reduced potential for producing cytochrome p-450 based drug interaction, report revealed that drug in both oral and IV formulation was effective for the treatment of GERD over 24 hours and was well tolerated in a variety of patients.
- **Basak.et.al25. (2002)** to prepared an enteric coated pancreatin tablets with five different enteric polymers. The tablets were evaluated on the basis of their disintegration time, loss on drying and enzyme activities periodically during a period of 90 days. All tablets but one stored at 45°C and 75% RH failed to meet disintegration test IP. Tablets coated with cellulose acetate phthalate and wincoat polymers failed disintegration test IP even at room temperature. All tablets formulations stored at 45°C and 75% RH failed loss on drying test IP. It was observed that the activities of pancreatic enzymes decreased on storage, but the loss in activity of individual enzymes were found to be different. Acrycoat L100 with polyvinylidene chloride packaging had shown promising results.

3.1 Stability Protocol Preparation of API

These Protocols provide definitions, compilation guidance, and other information to assist report preparers and to ensure consistency in the interpretation of the Performance Indicators. It contains the following elements:

- Information on batches tested (commercial formula)
- Container-closure system (commercial)

- Stability specifications (only for APIs)
- Analytical methods – stability indicating (cross-reference)
- Stability plan (schedule)
- Tabulated test results (including specifications)
- Analysis/discussion of data (statistical if negative trend)
- Re-test or shelf-life proposal (including storage condition)
- Post approval commitments

Information on batches tested (commercial formula) and Container-closure system (commercial)

Batches tested (tabulated format) should include below information:

- API batch number
- Manufacturing date
- Manufacturing site
- Batch size (in gm and in number of units)
- Container-closure (primary packaging)
- Date of initial analysis (release)

Table 4: API Batch data for stability testing

	1rd	3rd	6th
Batch number of API	SL/18/X/001	SL/18/X/002	X/S00074/074
Manufacturing date	Mar-18	Mar-18	Mar-18
Manufacturing site	Chemical research and development		
Batch size (gm.)	60.03 gm.	59.95 gm.	60.10 gm.
Batch type (full scale, pilot, etc.)	Lab- scale	Lab- scale	Lab- scale
Container closure System (Primary packaging)	Pack the sample in transparent polythene bag, twist and tie and covered with black polythene bag along with Silica pouch, fill with nitrogen gas; twist and tie. Further keep the above in triple laminated Aluminium bag with Silica pouch, fill with nitrogen gas & seal it, finally the packed sample placed in HDPE drum.		
Date initial analysis	June-18	June-18	June-18

3.2 Stability specifications (only for APIs):

Specification is a list of tests, reference to analytical procedures, and proposed acceptance criteria, is addressed in ICH Q16A and Q16B. Stability studies should include testing of those attributes (parameters) of the API that are susceptible to change during storage and thus are likely to influence quality, safety, and/or efficacy. The testing should cover, as appropriate, the physical, chemical, biological, and microbiological attributes.

Table 5: API Specification

S.No.	Test	Acceptance Criteria	Method Reference
1.	Appearance	White or almost white powder	EP
2.	Identification • By IR	The IR absorption spectrum obtained from the sample should be concordant with the IR spectrum of the working standard.	EP
3.	Water content by KF titrator (%w/w, use about 0.15g of sample)	Between 5.9 and 6.9	EP
4.	Related substances by HPLC (%w/w) a) Impurity-A b) Impurity-B c) Impurity-C d) Impurity-(D+F) e) Impurity-E Unspecified impurity Total impurity	 Not more than 0.2 Not more than 0.1 Not more than 0.1 Not more than 0.2 Not more than 0.1 Not more than 0.10 Not more than 0.5	EP
5.	Assay by potentiometry (% w/w, on anhydrous basis)	Not lass than 99.0 and not more than 101.0	EP

3.3 Stability indicating analytical methods:

Test methods may be reliable, meaningful, and specific. Accuracy, sensitivity, specificity, and reproducibility of test methods employed by the firm shall be established and documented. Testing procedures must include a stability indicating test which will distinguish the active ingredient from any degradation products and be able to make a reliable estimate of the quantity of any degradate. The stability indicating test does not have to be the assay method used to determine product strength.

Although specific methods are critical to determine product stability, they do not have to employ any specific technique. The use of quantitative analysis, where limits are known, such as thin layer chromatography, may be satisfactory. While many USP tests are specific for the drug or it's degrades and may be used for stability testing, some USP monographs do not incorporate stability indicating tests. Additionally, it may be unreasonable to expect a manufacturer to develop specific methodology for each component of some multi-component drugs containing ingredients of botanical origin such as benzoin, Peruvian balsam or Tolu balsam.

3.4 Standard testing procedure for stability testing of API:

3.4.1 Appearance:

Object- To determine the appearance (colour and physical state) of the sample (API).

Apparatus required- Petri dish, Spatula

Material required- Sample

Procedure - Take about 2.3 gm. of sample in a dry petridish, spread it with spatula and observe under suitable light for its colour and presence of extraneous matter. Record the observations and report the result.

Observation:

At initial, 1st month, 3rd month and 6th month analysis of X drug substance (Batch no. SL/18/X/001, SL/18/X/002, X/S00074/074) for Long term and Accelerated storage condition**.** The appearance of X drug substance is white or almost white powder.

Result: X drug substance is white or almost white powder at initial, 1st month, 3rd month and 6th month analysis for Long term and Accelerated storage condition.

3.4.2 Identification by IR spectrophotometer

Object - Identification of the sample by comparing with standard sample using IR spectrophotometer.

Equipment – IR spectrophotometer, Analytical Balance

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Apparatus required- Pestle, compressor, dies, Butter paper, Spatula

Material required- Standard sample, Test sample, KBr

Standard Preparation – Triturate about 2 to 3 mg of sample (drug substance) working standard with about 300 to 400 mg of KBr (Potassium bromide), previously dried at 105 ºC for 1 hour. Grind the mixture in Pestle and carefully spread it uniformly in a suitable die and compress at a pressure of about 800 Mpa. In compressor, now carefully place the disc obtained in a holder and transfer to the sample beam of the IR spectrophotometer.

Test Preparation – Triturate about 2 to 3 mg. of the sample with about 300 to 400 mg. of KBr (Potassium bromide) , previously dried at 105ºC for 1 hour. Grind the mixture in Pestle and carefully spread it uniformly in a suitable die & compress at a pressure of about 800 Mpa. In compressor, now carefully place the disc obtained in a holder and transfer to the sample beam of the IR spectrophotometer.

Principle: It involves the absorption of electromagnetic radiation in the infrared region of the spectrum which results in changes in the *vibrational energy* of molecule. Since, usually all molecules will be having vibrations in the form of stretching, bending, etc., the absorbed energy will be utilised in changing the energy levels associated with them. It is a valuable and formidable tool in identifying organic compounds which have polar chemical bonds (such as OH, NH, CH, etc.) with good charge separation (strong dipoles).

Procedure – Record the spectra of the standard preparation & test preparation over the range from about 4000 Cm^{-1} to 400 Cm^{-1}.

Interpretation – The absorption maxima in the spectrum obtained with the sample, correspond in same wave numbers and relative intensity to those in the spectrum obtained with the working standard. Record the observations & report the result.

Observation- At initial, 1st month, 3rd month and 6th month analysis for long term and accelerated storage condition the IR spectrum of X drug substance is concordant with the IR spectrum of working standard.

Result: The IR spectrum of X substance is concordant with the IR spectrum of working standard at initial, 1st month, 3rd month and 6th month analysis for long term and accelerated storage condition.

3.4.3 Water content (By KF auto titrator):

Object – To determine the water content (%w/w) in the given sample (API).

Equipment required – KF auto titrator, Analytical balance.

Material required – Karl Fischer reagent, Methanol, Sample**.**

Procedure - Transfer about 50 ml. of methanol to the titration vessel and neutralize the Methanol using Karl Fischer reagent. After the end point has reached, Weigh about 0.15 gm. of sample and transfer quickly into the titration vessel. Use a stirrer to dissolve the sample and then titrate it to the end point with Karl Fischer reagent.

Calculate the water content as follows:

Water content (%w/w) = $(V_{KFR} \times \text{KF factor} \times 100) / W_s \times 1000$

Where,

V_{KFR} = Volume of Karl Fischer reagent consumed.

W_s = Weight of the sample.

Observation: At initial, 1st month, 3rd month and 6th month analysis for long term and accelerated storage condition the water content of X drug substance is concordant with the water content of working standard.

Result: Water content of given X drug samples (Acceptance Criteria: Between 5.9 and 6.9)

S.No.	Batch No.	Initial	1st	3rd	6th
1.	SL/X/18/001 Long Term	6.50	6.26	6.37	6.59
	SL/X/18/002 Long Term	6.46	6.13	6.42	6.59
	X/S00074/074LongTerm	6.48	6.37	6.43	6.25
2.	SL/X/18/001 Long Term	6.50	6.43	6.40	6.42
	SL/X/18/002 Long Term	6.46	6.60	6.42	6.28
	X/S00074/074LongTerm	6.48	6.28	6.31	6.38

3.4.4 Related Substances – (By HPLC)

Object- To determine the related substances (Impurities) of given sample (API) by HPLC instrument.

Equipment required- HPLC, analytical balance, Ultra Sonicator, pH meter.

Apparatus required- 1 litre gradated plastic beaker, 1 litre graduated glass bottle (for Mobile phase preparation), measuring cylinder of 1000 ml. and 5mL,10mL,25mL,50mL and 100mL Volumetric flask, HPLC vials, Tissue paper, butter paper etc.

Reagents required-

- Dipotassium hydrogen phosphate : (AR grade)
- Orthophosphoric Acid : (AR grade)
- Acetonitrile :(HPLC grade)
- Sodium Hydroxide : (AR grade)
- Water : Milli-Q
- Standard sample
- Test sample
- Impurity-A: 5-(Difluoromethoxy)-2-[[(3,4-dimethoxypyridin-2yl)methyl]sulfonyl]-1-*H*-benzimidazole.
- Impurity-B: 5-(Difluoromethoxy)-2-[[(3,4-dimethoxypyridin-2yl)methyl]sulfanyl]-1-*H*-benzimidazole.
- Impurity-C: 5-(Difluoromethoxy)-1-*H*-benzimidazole-2-thiol.
- Impurity-D: 5-(Difluoromethoxy)-2-[(RS)[(3,4-dimethoxypyridin 2yl)methyl]sulfinyl]-1- methyl-1*H*-benzimidazole.
- Impurity-E: Mixture of the stereoisomers of the 6,6'-bis(difluoromethoxy)-2,2'-bis[[(3,4-dimethoxypyridin-2-yl)methyl]sufinyl]-1-*H*,1'*H*-5,5"-bibenzimidazolyl
- Impurity-F:6-(Difluoromethoxy)-2-[(RS)-[(3,4-dimethoxypyridin 2yl)methyl]sulfinyl]-1- methyl-1*H*-benzimidazole.

Principle: High performance Liquid Chromatography (HPLC) is an analytical process utilizing special instrument designed to separate, quantify and analyse component of a chemical mixture. Sample of interest are introduced to a solvent flow path, carried through a column packed with specialized materials for component data is obtained through the combination of a detection mechanism coupled with a data recording system. All this occur under pressure, which should not reach or exceed 15000psi.

Chromatography conditions – Liquid chromatograph, equipped with a variable wave length UV detector, an injector and a data processor.

Column : Hypersil ODS (125mm × 4.0 mm, 5.0µm)

Flow rate : 1.0 ml/minute

Wave length : 295nm and 305nm

Column temperature : 40°C

Sample temperature : 5°C

Injection volume : 20µL

Run time : 55 minutes

Retention time : The retention time of given sample (API) is about 8.5minutes.

Elution : Gradient mode

Gradient mode :

Time (Minutes)	0	40	45	55
% of MP-A	80	20	80	80
% of MP-B	20	80	20	20

Mobile phase-A : Dissolved 1.74g of dipotassium hydrogen phosphate in 1000mL of water. Shaken well to mix. Adjusted the pH to 7.00 ± 0.05 with diluted orthophosphoric acid.

Mobile phase-B : Acetonitrile.

Diluents : Mixture of Acetonitrile and sodium hydroxide solution (40mg/L) in the ratio of 1:1 (%v/v). Shaken well to mix.

Preparation:

Reference solution (b):

Dissolve the contents of vial of X for system suitability CRS (Containing impurities A, B, C, D and E) in 1mL of diluent.

(Or)

System Suitability Solution:

Accurately weigh about 2.5mg of X peak identification mixture and transfer into a 5mL volumetric flask. Add 3mL of diluent to it, dissolve and dilute to volume with diluent. Shake well to mix.

(Or)

Impurity Stock Solution-1:

Accurately weigh about 2.5mg of each of impurity-B, impurity-C and impurity-E reference/working standards and transfer into a 25mL volumetric flask. Add 10mL of diluent to it, dissolve and dilute to volume with diluent. Shake well to mix.

Impurity Stock Solution-2:

Accurately weigh about 2.5mg of each of impurity-A and impurity-(D+F) reference/working standards and transfer into a 25mL volumetric flask. Add 10mL of diluent to it, dissolve and dilute to volume with diluent. Shake well to mix.

System Suitability Stock Solution:

Transfer 5mL of the impurity stock solution-1 and 10mL of the impurity stock solution-2 into a 100mL volumetric flask and dilute to volume with diluent. Shake well mix.

System Suitability Solution:

Accurately weigh about 23mg of X reference/working standard and transfer into a 50mL volumetric flask. Add about 5mL of system suitability stock solution and then add 10mL of diluent to it, sonicate to dissolve and dilute to volume with diluent. Shake well to mix.

Reference Solution (a):

Accurately weigh about 23mg of X reference/working standard and transfer into a 50mL volumetric flask. Add about 5mL of diluent to it, dissolve and dilute to volume with diluent. Shake well to mix.

Transfer 1mL of the above solution into a 100mL volumetric flask, dilute to volume with diluent and shake well to mix.

Further transfer 1mL of the above solution into a 10mL volumetric flask, dilute to volume with diluent and shake well to mix.

Sample Solution:

Accurately weigh about 23mg of sample and transfer into a 50mL volumetric flask. Add 20mL of diluent to it, dissolve and dilute to volume with diluent. Shake well to mix.

Procedure:

Equilibrate the system for 30 minutes. Run the system as per the injection sequence given. Record the chromatograms and measure the peak responses. Disregard the peak due to blank. Disregard the peaks with area less than 0.05% w/w. Integrate reference solution (a) and reference solution (b) at 290nm. Integrate the sample solution at 305 nm to report Impurity-C and at 290nm to report rest of the impurities.

Injection Sequence

Name of the solution	**No. of injection**
Blank	1
Reference solution(b)/ System suitability solution	1
Blank	1
Reference solution(a)	6
Sample solution	1
Reference solution(b)/ System suitability solution- Bracketing	1

System Suitability Criteria:

1. The USP resolution between Impurity-(D+F) peak and Impurity-E peak as obtained from the reference solution (b)/ system suitability solution should not be less than 1.5.
2. The USP tailing factor of Pantoprazole peak as obtained from the reference solution(b)/system suitability solution should be not more than 2.0.
3. The %RSD of area of Pantoprazole peak as obtained from six replicate injections of reference solution (a) should be not more than 5.0.

The Relative Retention Time (RRT) of the impurities is tabulated as below:

Name of the component	RRT at 305nm	RRT at 290nm
Impurity-C	~0.6	-
Impurity-A	-	~0.9
X	-	1.00
Impurity-(D+F)	-	~1.2
Impurity-E	-	~1.3
Impurity-B	-	~1.5

For calculation of content, multiply the peak area of Impurity-C with corresponding correction factor; Impurity-C-0.3.

Note: The column has to be washed well as sample is basic in nature and may spoil the column in case of poor washing.

Column washing:

Wash the column for 1hour after completion of analysis Mix water, acetonitrile and formic acid in the ration 1:1:0.05 (%v/v/v). Shake well to mix.

Calculation

$$\text{Known Impurities at 290 nm (\% w/w)} = \frac{A_I}{A_S} \times \frac{W_S}{50} \times \frac{1}{100} \times \frac{1}{10} \times \frac{50}{W_T} \times P$$

$$\text{Impurity} - \text{C at 305 nm (\% w/w)} = \frac{A_{I\,Imp-C}}{A_S} \times \frac{W_S}{50} \times \frac{1}{100} \times \frac{1}{10} \times \frac{50}{W_T} \times P$$

$$\text{Unspecified impurity at 290 nm (\%w/w)} = \frac{A_{USI}}{A_S} \times \frac{W_S}{50} \times \frac{1}{100} \times \frac{1}{10} \times \frac{50}{W_T} \times P$$

$$\text{Total unknown Impurities at 290 nm (\% w/w)} = \frac{A_{TUI}}{A_S} \times \frac{W_S}{50} \times \frac{1}{100} \times \frac{1}{10} \times \frac{50}{W_T} \times P$$

$$\text{Total known impurities (\% w/w)} = \text{Known impurities at 290 nm (\% w/w)} + \text{Impurity} - \text{C at 305 nm (\% w/w)}$$

$$\text{Total impurities (\% w/w)} = \text{Total known impurities (\% w/w)} + \text{Total unknown impurities (\% w/w)}$$

Where,

A_I : Area of respective impurity peak in the sample solution at 290 nm.

$A_{I\ Imp\text{-}C}$: Area of impurity-C peak in the sample solution at 305 nm.

A_S : Average area of X peak in six replicate injections of reference solution at 290 nm

W_S : Weight of X reference/ working standard in reference solution (a) (mg).

W_T : Weight of sample in mg.

P : Purity of X reference/ working standard.

AUSI : Area of unspecified impurity in sample solution at 290 nm.

ATUI : Area of total unknown impurities in sample solution at 290 nm.

RRF : Relative response factor.

3.4.5 Observation-initial analysis of X drug substance:

Sample information	
Sample name- Blank at 290nm Test parameter-Related substance	Date of initial analysis : June-18

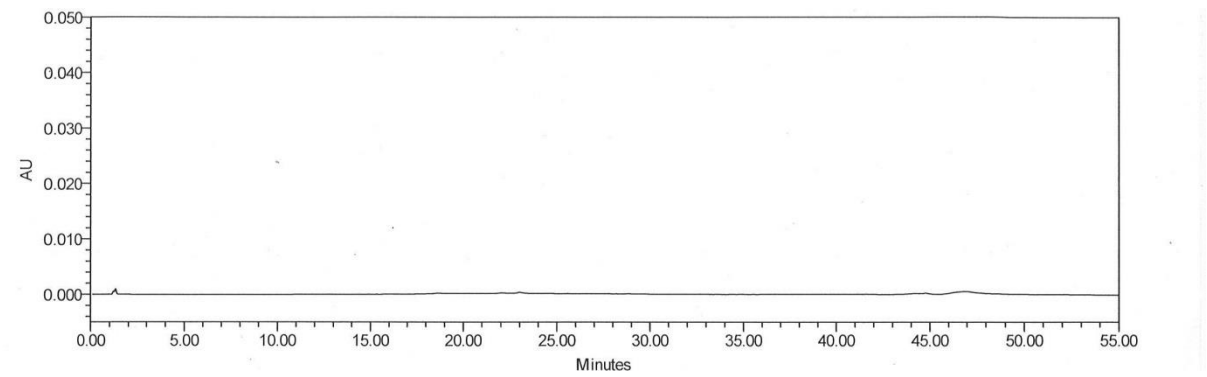


Figure: 1 Typical chromatogram blank in the test for related substances at 290 nm

Sample information	
Sample name- Blank at 305nm Test parameter-Related substance	Date of initial analysis : June-18

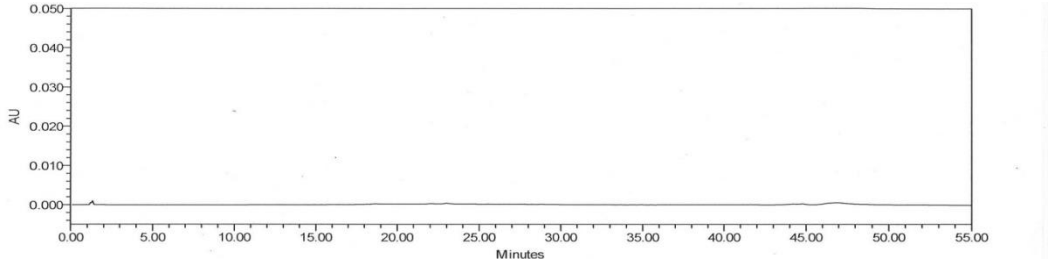
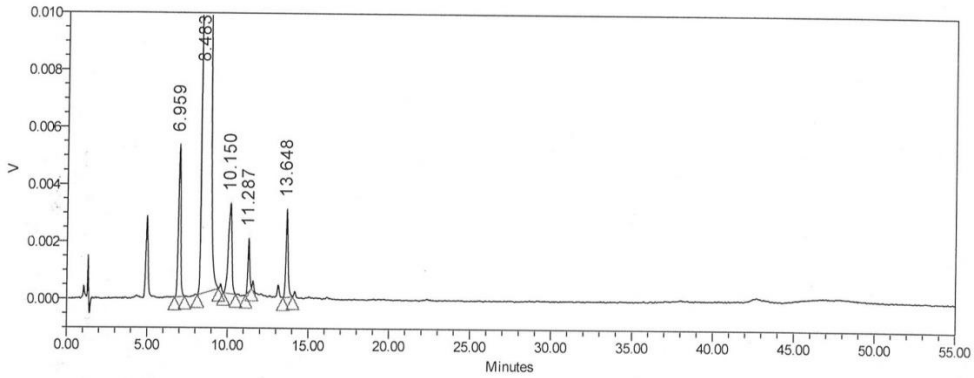


Figure: 2 Typical chromatogram of blank in the test for related substances at 305 nm.

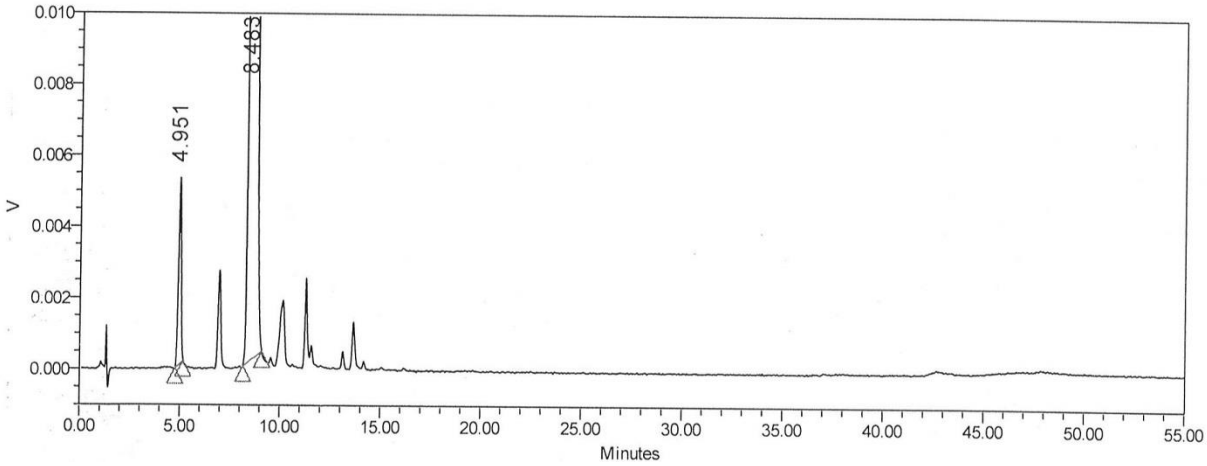
Sample Information	
Sample Name: System Suitability Solution at 290nm Test Parameter: Related substances	Date of manufacturing: Mar-18 Date of initial analysis: June-18



	Peak Name	RT	Height (µV)	Area	% Area	Int Type	RT Ratio	USP Resolution	USP Plate Count	USP Tailing
1	IMP-A	6.96	5322	53910	0.26	BB	0.82		11202	1.12
2	PAN	8.48	2212497	20610005	99.29	BB	1.00	6.00	20366	1.18
3	IMP-D+F	10.15	3146	48055	0.23	BB	1.20	5.40	11663	0.86
4	IMP-E	11.29	1861	15073	0.07	BB	1.33	3.81	42897	1.09
5	IMP-B	13.65	3079	30918	0.15	BB	1.61	9.76	45921	1.23
Sum				20757961						

Figure: 3 Typical chromatogram of system suitability solution in the test for related substances at 290 nm.

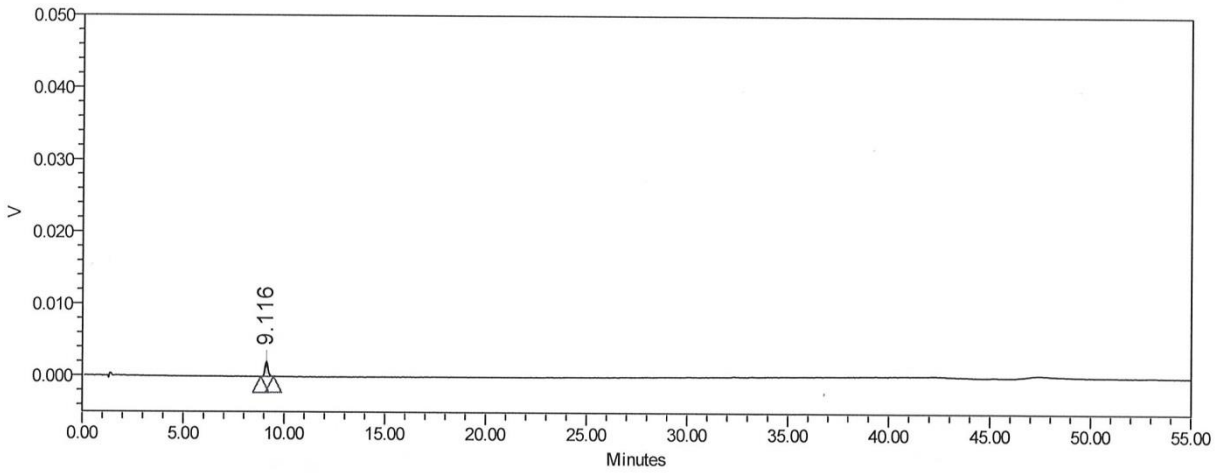
Sample Information	
Sample Name: System Suitability Solution at 305nm Test Parameter: Related substances	Date of manufacturing: Mar-18 Date of initial analysis: June-18



	Peak Name	RT	Height (µV)	Area	% Area	Int Type	RT Ratio	USP Resolution	USP Plate Count	USP Tailing
1	IMP-C	4.95	5229	49781	0.57	BB	0.58		6057	1.07
2	PAN	8.48	906841	8691716	99.43	BB	1.00	13.95	19317	1.19
Sum				8741497						

Figure: 4 Typical chromatogram of system suitability solution in the test for related substances at 305 nm.

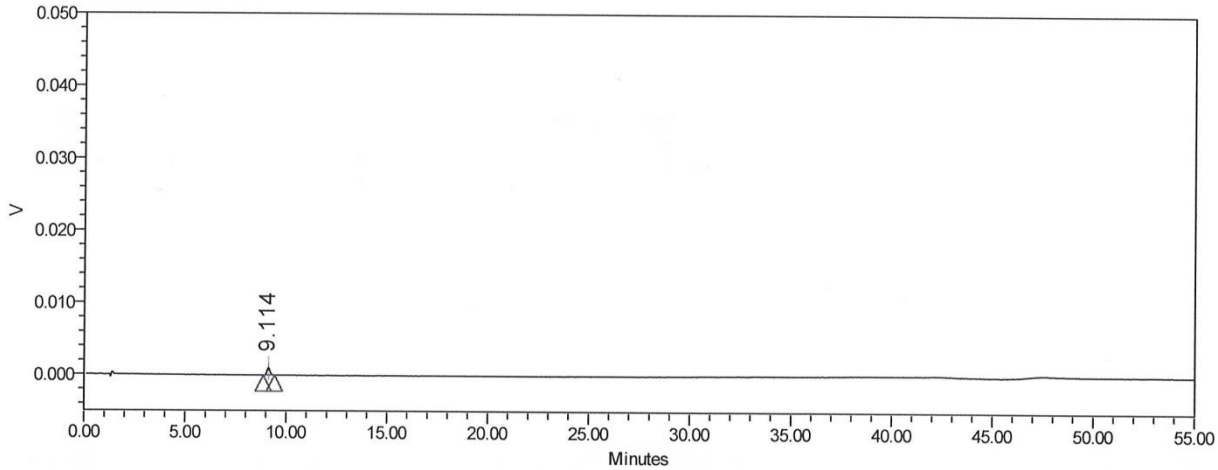
Sample Information	
Sample Name: Reference solution (a) at 290nm Test Parameter: Related substances	Date of manufacturing: Mar-18 Date of initial analysis: June-18



	Peak Name	RT	Height (µV)	Area	% Area	Int Type
1	PAN	9.12	2054	21410	100.000	BB
Sum				21410		

Figure: 5 Typical chromatogram of reference solution (a) in the test for related substances at 290 nm.

Sample Information	
Sample Name: Reference solution (a) at 305nm Test Parameter: Related substances	Date of manufacturing: Mar-18 Date of initial analysis: June-18



	Peak Name	RT	Height (µV)	Area	% Area	Int Type
1	PAN	9.11	997	10423	100.000	BB
Sum				10423		

Figure: 6 Typical chromatogram of reference solution (a) in the test for related substances at 305 nm.

3.4.6 Observation stability analysis of X drug substance:

Sample information	
Sample name-Blank at 290nm Test parameter-Related substance	 Stability Analysis

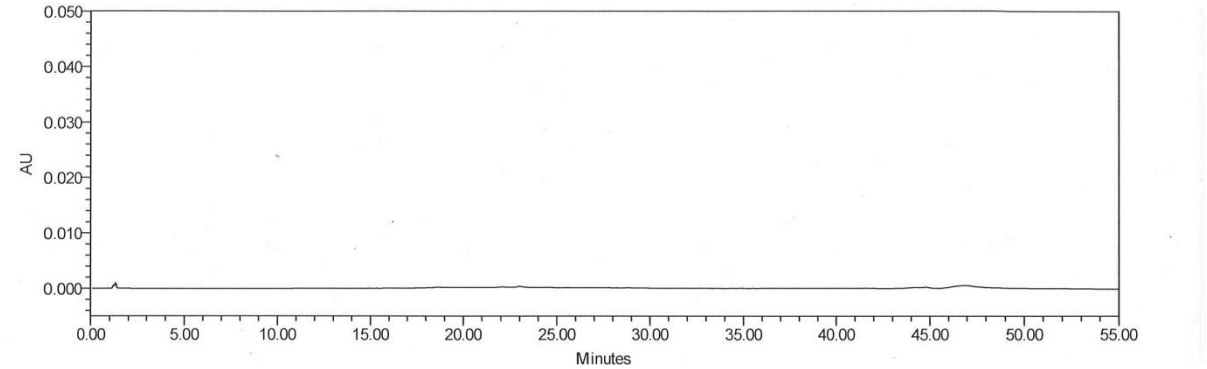


Figure: 7 Typical chromatogram blank in the test for related substances at 290 nm

Sample information	
Sample name-Blank at 305nm Test parameter-Related substance	 Stability Analysis

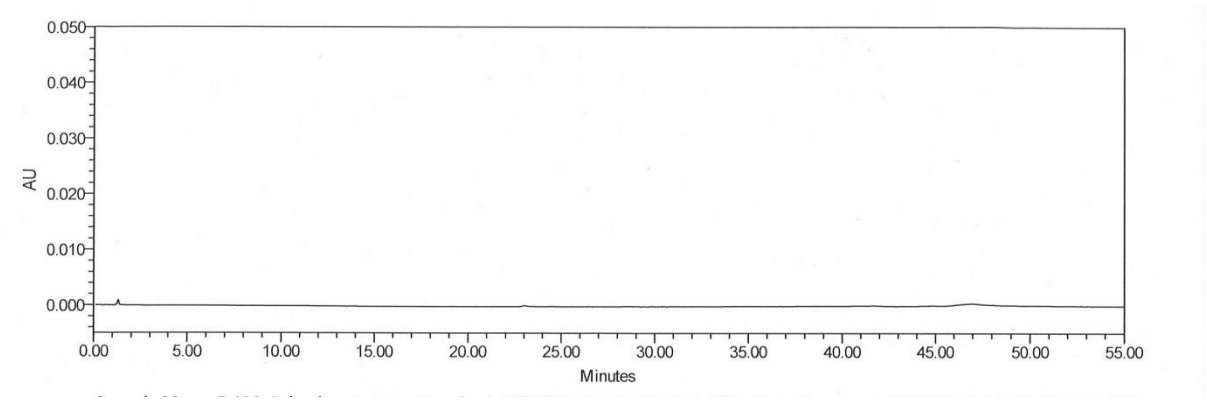
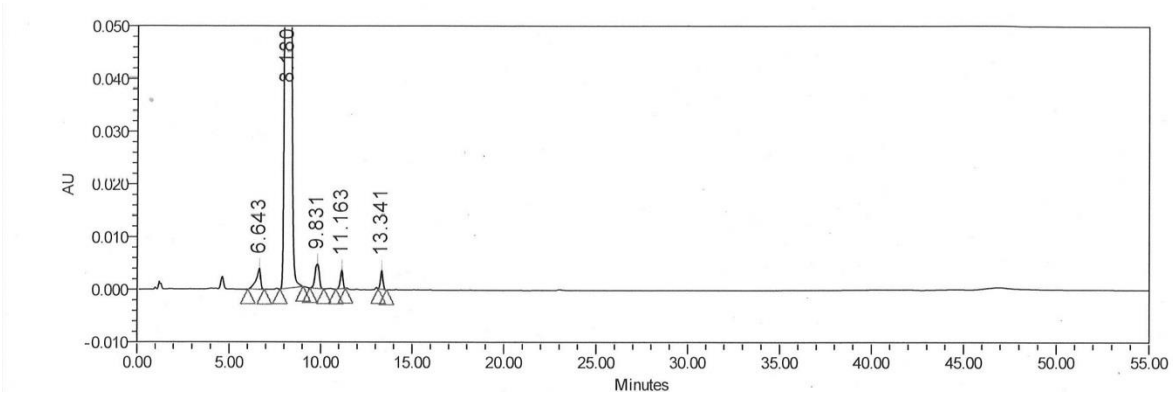


Figure: 8 Typical chromatogram blank in the test for related substances at 305 nm

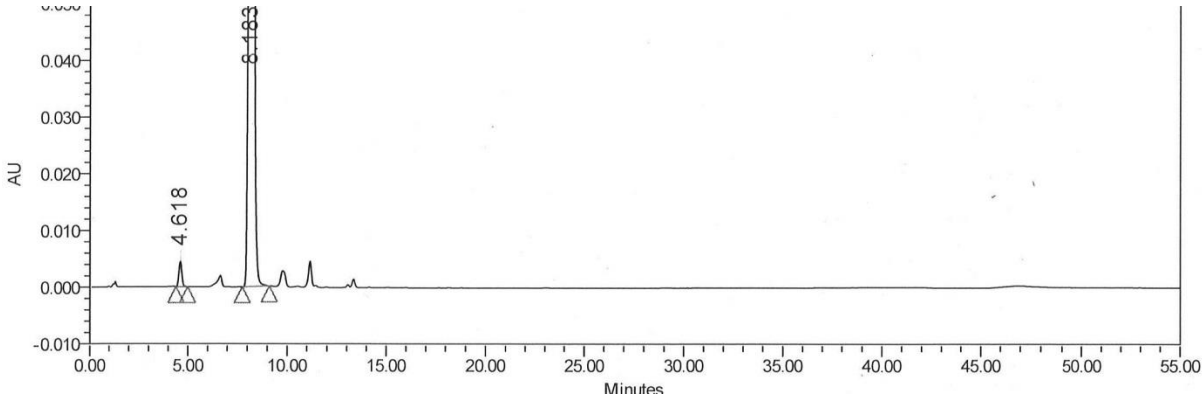
Sample Information	
Sample Name: System Suitability Solution at 290nm Test Parameter: Related substances	Date of manufacturing: Mar-18 Stability Analysis



	Peak Name	RT	Height (µV)	Area	% Area	Int Type	RT Ratio	USP Resolution	USP Tailing	USP Plate Count
1	Impurity-A	6.64	3936	66293	0.260	BB	0.81		0.66	3473
2	Pantoprazole	8.18	2280441	25266236	99.179	BB	1.00	4.02	0.96	12702
3	Impurity-(D+F)	9.83	4601	73904	0.290	BB	1.20	4.87	0.86	10518
4	Impurity-E	11.16	3492	35720	0.140	BB	1.36	4.03	0.81	27356
5	Impurity-B	13.34	3532	33183	0.130	BB	1.63	8.19	1.08	45177
Sum				25475336						

Figure: 9 Typical chromatogram Reference Solution(b) in the test for related substances at 209 nm

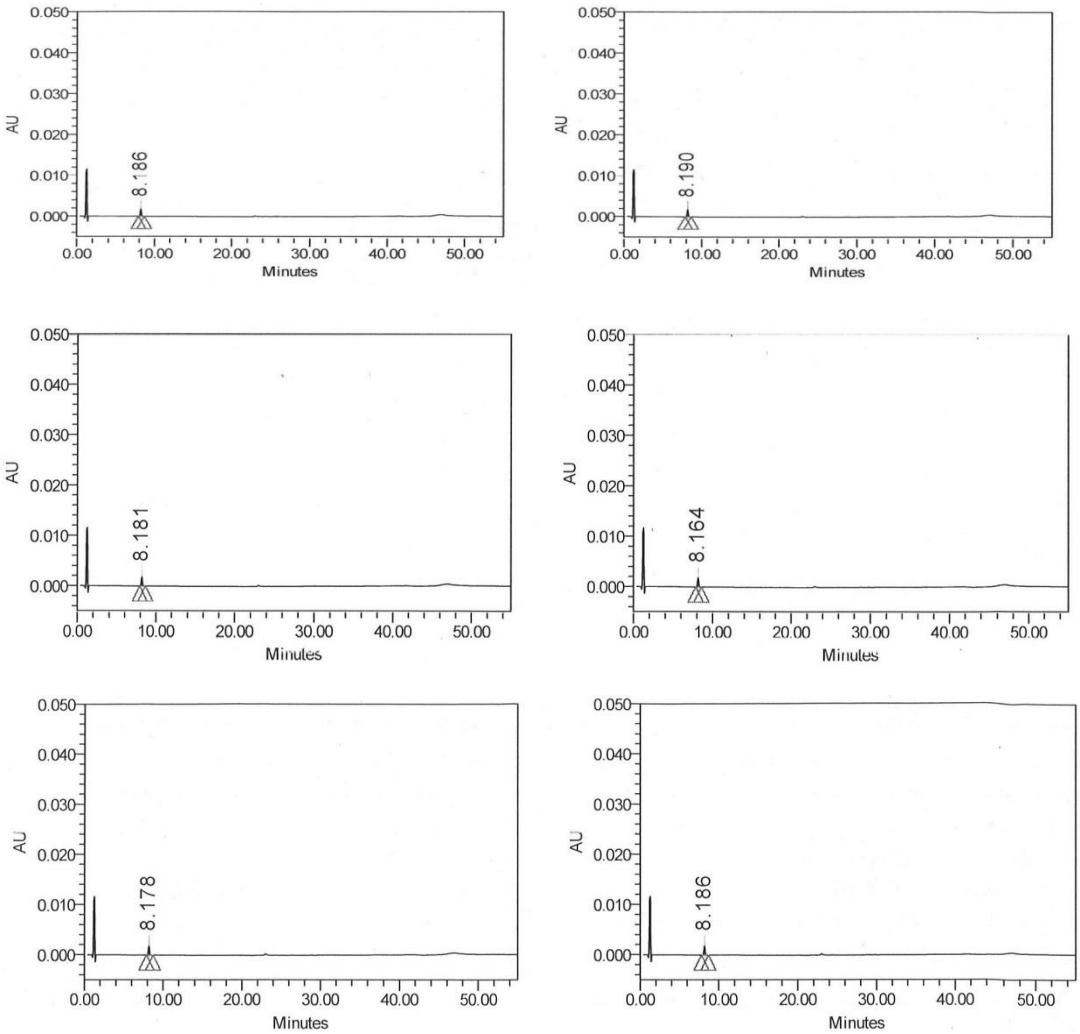
Sample Information	
Sample Name: System Suitability Solution at 305nm Test Parameter: Related substances	Date of manufacturing: Mar-18 Stability Analysis



	Peak Name	RT	Height (µV)	Area	% Area	Int Type	RT Ratio	USP Resolution	USP Tailing	USP Plate Count
1	Impurity-C	4.62	4406	45776	0.442	BB	0.56		1.03	4526
2	Pantoprazole	8.18	917762	10301819	99.558	BB	1.00	12.46	0.96	12338
Sum				10347595						

Figure: 10 Typical chromatogram Reference Solution(b) in the test for related substances at 305 nm

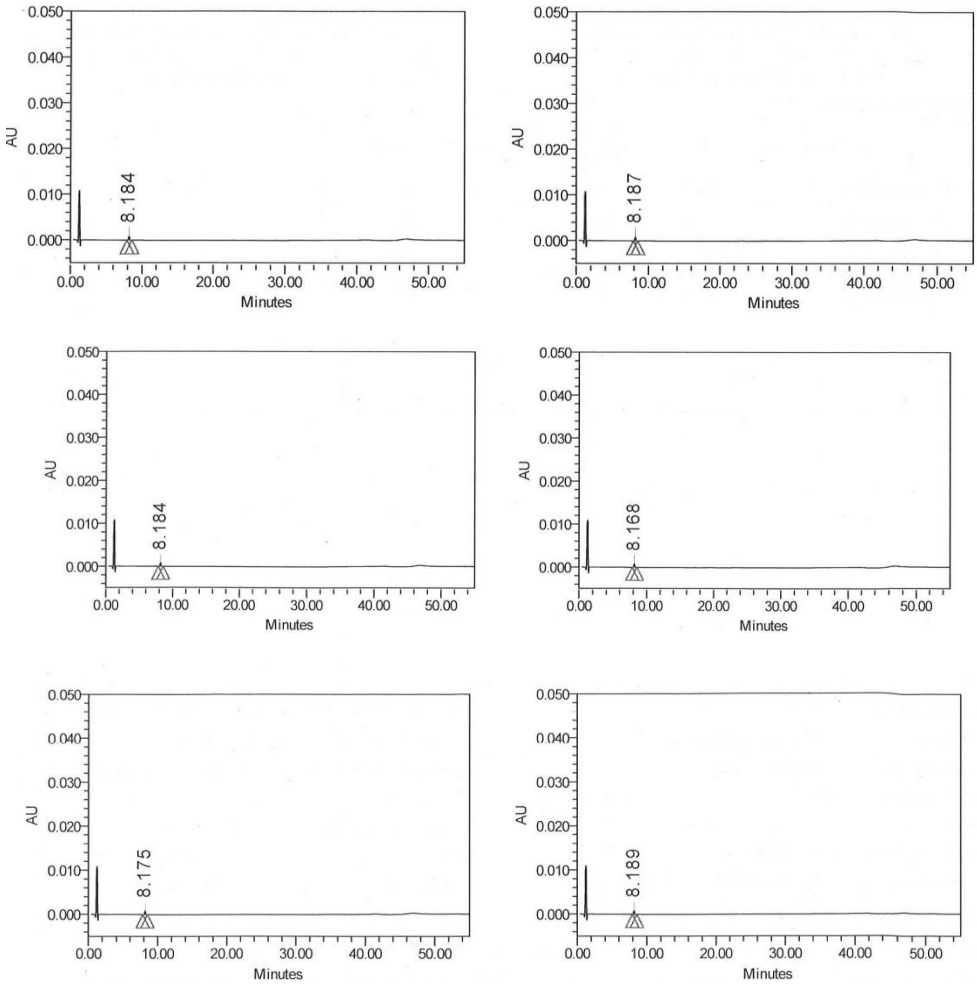
Sample Information	
Sample Name: Reference solution (a) at 290nm Test Parameter: Related substances	Date of manufacturing: Mar-18 Stability Analysis



	Peak Name	RT	Height (µV)	Area	Int Type
1	PAN	8.16	1760	18135	BB
2	PAN	8.18	1763	18179	BB
3	PAN	8.18	1767	18271	BB
4	PAN	8.19	1769	18304	BB
5	PAN	8.19	1767	18088	BB
6	PAN	8.19	1761	18154	BB
Mean				18188	
Std. Dev.				83.24	
% RSD				0.46	

Figure: 11 Typical chromatogram Reference Solution (a) in the test for related substances at 290 nm

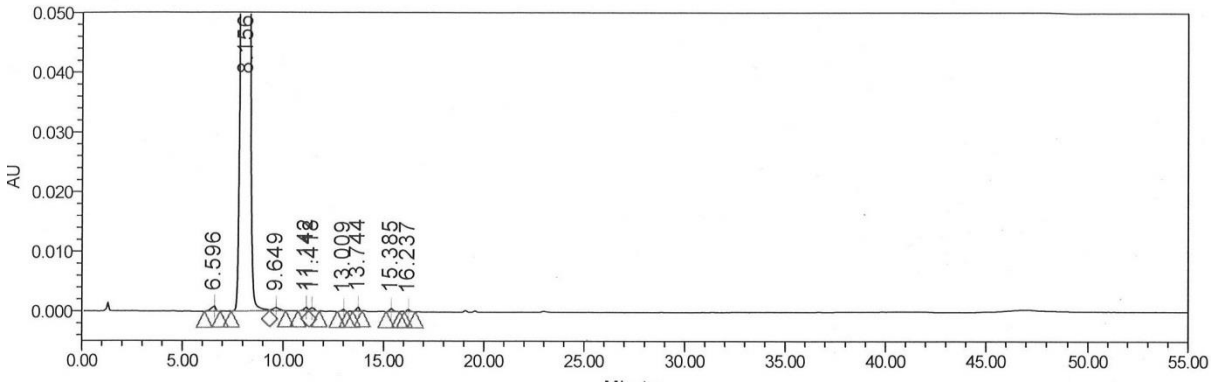
Sample Information	
Sample Name: Reference solution (a) at 305nm Test Parameter: Related substances	Date of manufacturing: Mar-18 Stability Analysis



	Peak Name	RT	Height (µV)	Area	Int Type
1	Pantoprazole	8.17	765	7940	BB
2	Pantoprazole	8.18	761	7825	BB
3	Pantoprazole	8.18	762	7775	BB
4	Pantoprazole	8.18	766	7964	BB
5	Pantoprazole	8.19	764	7892	BB
6	Pantoprazole	8.19	766	7866	BB
Mean				7877	
Std. Dev.				70.59	
% RSD				0.90	

Figure: 12 Typical chromatogram Reference Solution (a) in the test for related substances at 305 nm

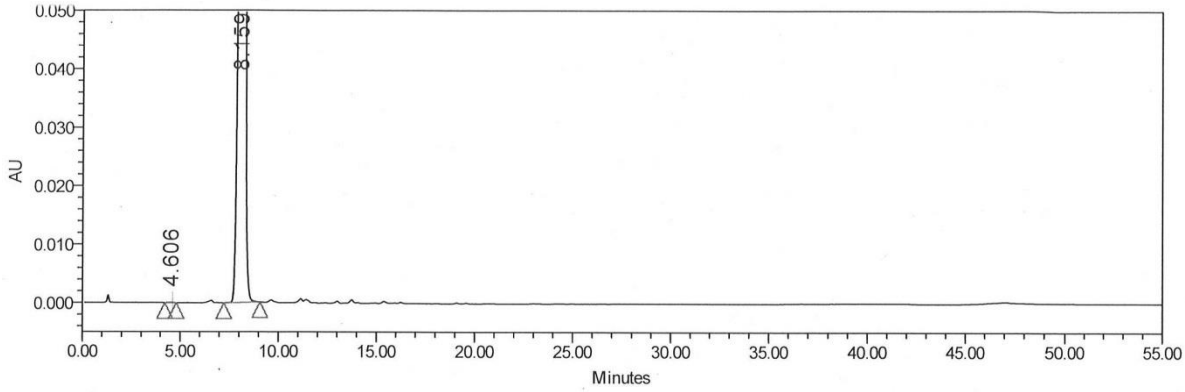
Sample Information	
Sample Name: X long term sample at 290nm Test Parameter: Related substances	Date of manufacturing: Mar-18 Stability Analysis



	Peak Name	RT	Height (µV)	Area	% Area	Int Type	RT Ratio
1	Impurity-A	6.60	818	13190	0.069	BB	0.81
2	Pantoprazole	8.16	1269063	19133095	99.675	BV	1.00
3	~~Impurity-(D+F)~~	9.65	545	12570	0.065	VB	1.18
4	Impurity-E	11.14	619	8243	0.043	BV	1.37
5	Peak5	11.42	516	8462	0.044	VB	1.40
6	Peak6	13.01	311	3385	0.018	BB	1.60
7	Peak7	13.74	610	6584	0.034	BB	1.69
8	Peak8	15.39	472	6081	0.032	BB	1.89
9	Peak9	16.24	335	3944	0.021	BB	1.99
Sum				19195553			

Figure: 13 Typical chromatogram Sample solution in the test for related substances at 290 nm

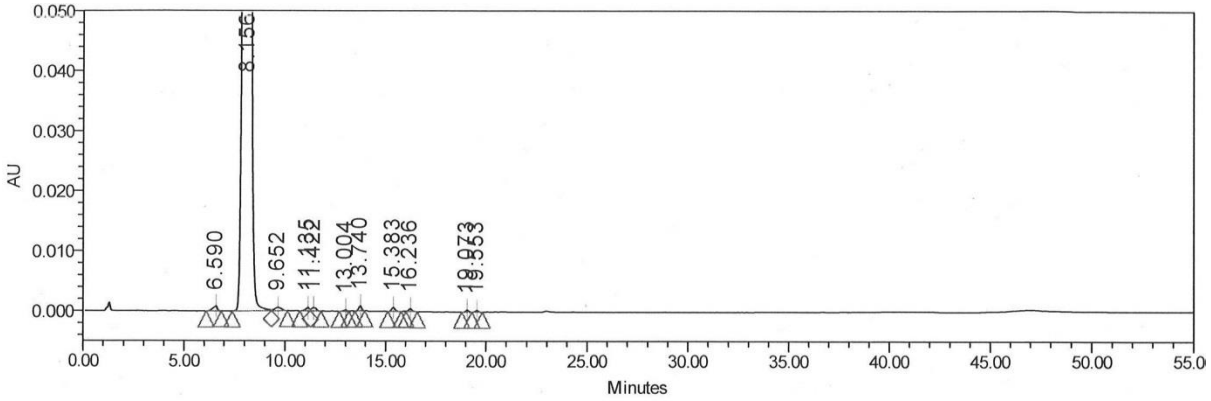
Sample Information	
Sample Name: X long term sample at 305nm Test Parameter: Related substances	Date of manufacturing: Mar-18 Stability Analysis



	Peak Name	RT	Height (µV)	Area	% Area	Int Type	RT Ratio
1	Impurity-C	4.61	54	850	0.011	BB	0.56
2	Pantoprazole	8.16	520794	7956760	99.989	BB	1.00
Sum				7957610			

Figure: 14 Typical chromatogram sample solution in the test for related substances at 305 nm

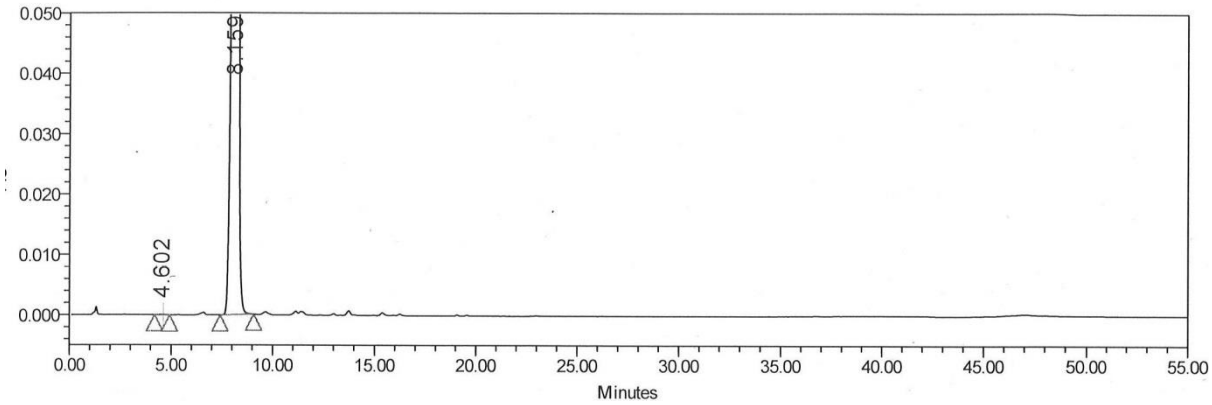
Sample Information	
Sample Name: X accelerated sample at 290nm Test Parameter: Related substances	Date of manufacturing: Mar-18 Stability Analysis



	Peak Name	RT	Height (µV)	Area	% Area	Int Type	RT Ratio
1	Impurity-A	6.59	833	13359	0.069	BB	0.81
2	Pantoprazole	8.16	1279886	19347008	99.612	BV	1.00
3	Impurity-(D+F)	9.65	583	13535	0.070	VB	1.18
4	Impurity-E	11.14	566	7497	0.039	BV	1.37
5	Peak5	11.42	553	9620	0.050	VB	1.40
6	Peak6	13.00	245	2629	0.014	BB	1.59
7	Peak7	13.74	812	8877	0.046	BB	1.68
8	Peak8	15.38	666	8421	0.043	BB	1.89
9	Peak9	16.24	463	5369	0.028	BB	1.99
10	Peak10	19.07	339	3371	0.017	BB	2.34
11	Peak11	19.55	264	2649	0.014	BB	2.40

Figure: 15 Typical chromatogram Sample solution in the test for related substances at 290 nm

Sample Information	
Sample Name: X accelerated sample at 305nm Test Parameter: Related substances	Date of manufacturing: Mar-18 Stability Analysis



	Peak Name	RT	Height (µV)	Area	% Area	Int Type	RT Ratio
1	Impurity-C	4.60	105	1735	0.022	BB	0.56
2	Pantoprazole	8.16	525810	8058469	99.978	BB	1.00
Sum				8060204			

Figure: 16 Typical chromatogram Sample solution in the test for related substances at 305 nm

3.4.5. Assay by Potentiometry:

Object – To determine the assay by potentiometry in the given sample (API).

Equipment required – KF auto titrator, Analytical balance, Titration vessel.

Instrument Parameters of Autotitrator MET U (Make: Metrohm)

General/Hardware

Device name	:	905_1
Device type	:	905 Titrando
Dosing device	:	4
Solution	:	0.1 N perchloric acid
Sensor	:	iSolvotrode 2
Stirring rate	:	4

Start conditions

Signal drift	:	off mV/min
Start volume	:	6 mL (0 mL for blank)
Dosing rate	:	Maximum (mL/min)
Start measured value - Dosing rate	:	Maximum (mL/min)
Start slope - Dosing rate	:	Maximum (mL/min)
Pause	:	10 sec

Titration parameters

Titration rate	:	User
Signal drift	:	50.0 mV/min
Minimum waiting time	:	0 sec
Maximum waiting time	:	26 sec
Dosing of increments-Volume increment	:	0.1 mL (0.01 mL for blank)
Dosing of increments-Dosing rate	:	Maximum mL/min
Temperature	:	25 ºC

Stop conditions

Stop volume	:	30 mL (1 mL for blank)
Stop measured value	:	Off mV
Stop EP	:	1
Volume after EP	:	1 mL
Filling rate	:	Maximum (mL/min)
EP criterion	:	30 mV
EP recognition	:	Greatest

Reagents and chemicals used

1. 0.1 N Perchloric acid standard solution
2. Glacial acetic acid
3. Acetic anhydride

Procedure

Accurately weigh and transfer about 0.2 g of the sample into a titration vessel, add 80 mL of glacial acetic acid and add 5 mL of acetic anhydride to it. Sonicate to dissolve and shake well to mix. Titrate the solution against 0.1 N perchloric acid standard potentiometrically. Perform the analysis in duplicate.

Perform the blank titration and report the average result by applying the blank correction.

Calculation

Calculate the assay by using the following formula

$$\text{Assay (\% w/w, on anhydrous basis)} = \frac{(V_S - V_B) \times N \times 0.2027 \times 100}{W} \times \frac{100}{100 - WC}$$

Where,

V_S : Volume of titrant consumed for sample determination

V_B : Volume of titrant consumed for blank determination

N : Normality of 0.1 N perchloric acid standard solutions

W : Weight of sample taken in g

WC : Water content of sample

Result: Assay by Potentiometry of given X drug samples (Acceptance Criteria: Not less than 99.0 and not more than 101.0)

S.No.	Batch No.	Initial	1st	3rd	6th
1.	SL/X/18/001 Long Term	99.7	99.8	99.9	100.1
	SL/X/18/002 Long Term	100.2	99.4	100.1	99.7
	X/S00074/074LongTerm	100.8	99.3	100.1	100.1
2.	SL/X/18/001 Long Term	99.7	99.6	99.8	100.4
	SL/X/18/002 Long Term	100.2	99.1	99.1	100.6
	X/S00074/074LongTerm	100.8	99.9	99.9	100.1

4. Results and Discussions

Long term and Accelerate condition stability analysis for initial, first month, third month and sixth month data:

Stability study report No.	X/10/09	Reference protocol No.	X/10	
Name of the Product	X	Date of Mfg.	Mar-18	
Batch No.	SL/18/X/001	Date Kept for stability	18/06/18	
Specification No.	X/EP/04	STP/PTP No.	X/EP/04	
Storage conditions	Pack the sample under inert atmosphere in transparent polyethylene, twist and tie, followed by black polyethylene bag, twist and tie and then place in a HDPE container.			
	Temperature: 30 ± 2 °C		Relative Humidity : 65 ± 5%	

S. No.	Tests	Acceptance criteria	Date of Analysis	Date of Analysis	Date of Analysis	Date of Analysis
			14/06/18	26/07/18	05/10/18	23/01/19
			Initial	1st month	3rd month	6th month
1	Appearance	White or almost white powder	Almost white powder	Almost white powder	Almost white powder	Almost white powder
2	Identification by IR	The IR spectrum of the sample should be concordant with the IR spectrum of Pantoprazole sodium sesquihydrate reference/ working standard.	Complies	Complies	Complies	Complies
3	Water by KF titrator (% w/w, use about 0.15 g of sample)	Between 5.9 and 6.9	6.50	6.26	6.37	6.59
4	Related substances by HPLC (% w/w) a) Impurity-A b) Impurity-B c) Impurity-C d) Impurity-(D+F) e) Impurity-E Unspecified impurity Total impurities	 Not more than 0.2 Not more than 0.1 Not more than 0.1 Not more than 0.2 Not more than 0.1 Not more than 0.10 Not more than 0.5	 0.06 Not detected Not detected Not detected 0.03 (BDL) 0.01 (BDL) 0.06	 0.08 BDL BDL BDL BDL BDL 0.08	 0.07 BDL BDL BDL BDL BDL 0.07	 0.08 BDL BDL BDL 0.05 BDL 0.13
5	Assay by Potentiometry(% w/w, on anhydrous basis)	*Not less than 99.0 and not more than 101.0*	99.7	99.8	99.9	100.1

Stability study report No.	X/10/09	**Reference protocol No.**	X/10	
Name of the Product	X	**Date of Mfg.**	Mar-18	
Batch No.	SL/18/X/002	**Date Kept for stability**	18/06/18	
Specification No.	X/EP/04	**STP/PTP No.**	X/EP/04	
Storage conditions	Pack the sample under inert atmosphere in transparent polyethylene, twist and tie, followed by black polyethylene bag, twist and tie and then place in a HDPE container.			
	Temperature: 30 ± 2 °C		**Relative Humidity : 65 ± 5%**	

S. No.	Tests	Acceptance criteria	Date of Analysis	Date of Analysis	Date of Analysis	Date of Analysis
			14/06/18	26/07/18	05/10/18	23/01/19
			Initial	1st month	3rd month	6th month
1	Appearance	White or almost white powder	Almost white powder	Almost white powder	Almost white powder	Almost white powder
2	Identification by IR	The IR spectrum of the sample should be concordant with the IR spectrum of Pantoprazole sodium sesquihydrate reference/ working standard.	Complies	Complies	Complies	Complies
3	Water by KF titrator (% w/w, use about 0.15 g of sample)	Between 5.9 and 6.9	6.46	6.13	6.42	6.59

S. No.	Tests	Acceptance criteria	Date of Analysis	Date of Analysis	Date of Analysis	Date of Analysis
			14/06/18	26/07/18	05/10/18	23/01/19
			Initial	1st month	3rd month	6th month
4	Related substances by HPLC (% w/w) a) Impurity-A b) Impurity-B c) Impurity-C d) Impurity-(D+F) e) Impurity-E Unspecified impurity Total impurities	 Not more than 0.2 Not more than 0.1 Not more than 0.1 Not more than 0.2 Not more than 0.1 Not more than 0.10 Not more than 0.5	 0.09 Not detected Not detected Not detected 0.03 (BDL) 0.02 (BDL) 0.09	 0.11 BDL BDL BDL BDL BDL BDL 0.11	 0.11 BDL BDL BDL BDL BDL BDL 0.11	 0.12 BDL BDL BDL BDL 0.05 0.05 0.22
5	Assay by Potentiometry(% w/w, on anhydrous basis)	*Not less than 99.0 and not more than 101.0*	100.2	99.4	100.1	99.7

Stability study report No.	X/10/09	**Reference protocol No.**	X/10	
Name of the Product	X	**Date of Mfg.**	Mar-18	
Batch No.	X/S00074/074	**Date Kept for stability**	18/06/18	
Specification No.	X/EP/04	**STP/PTP No.**	X/EP/04	
Storage conditions	Pack the sample under inert atmosphere in transparent polyethylene, twist and tie, followed by black polyethylene bag, twist and tie and then place in a HDPE container.			
	Temperature: 30 ± 2 °C		**Relative Humidity : 65 ± 5%**	

S. No.	Tests	Acceptance criteria	Date of Analysis	Date of Analysis	Date of Analysis	Date of Analysis
			14/06/18	26/07/18	05/10/18	23/01/19
			Initial	1st month	3rd month	6th month
1	Appearance	White or almost white powder	Almost white powder	Almost white powder	Almost white powder	Almost white powder
2	Identification by IR	The IR spectrum of the sample should be concordant with the IR spectrum of Pantoprazole sodium sesquihydrate reference/ working standard.	Complies	Complies	Complies	Complies
3	Water by KF titrator (% w/w, use about 0.15 g of sample)	Between 5.9 and 6.9	6.48	6.37	6.43	6.25
4	Related substances by HPLC (% w/w) a) Impurity-A b) Impurity-B c) Impurity-C d) Impurity-(D+F) e) Impurity-E Unspecified impurity Total impurities	 Not more than 0.2 Not more than 0.1 Not more than 0.1 Not more than 0.2 Not more than 0.1 Not more than 0.10 Not more than 0.5	 0.06 Not detected Not detected Not detected 0.04 (BDL) 0.02 (BDL) 0.06	 0.09 BDL BDL BDL BDL BDL BDL 0.09	 0.07 BDL BDL BDL BDL BDL BDL 0.07	 0.07 BDL BDL BDL BDL BDL 0.06 0.13
5	Assay by Potentiometry(% w/w, on anhydrous basis)	*Not less than 99.0 and not more than 101.0*	100.8	99.3	100.1	100.1

Stability study report No.	X/10/06	**Reference protocol No.**	X/10	
Name of the Product	X	**Date of Mfg.**	Mar-18	
Batch No.	SL/18/X/001	**Date Kept for stability**	18/06/18	
Specification No.	X/EP/04	**STP/PTP No.**	X/EP/04	
Storage conditions	Pack the sample under inert atmosphere in transparent polyethylene, twist and tie, followed by black polyethylene bag, twist and tie and then place in a HDPE container.			
	Temperature: 40 ± 2 °C		**Relative Humidity : 75 ± 5%**	

S. No.	Tests	Acceptance criteria	Date of Analysis	Date of Analysis	Date of Analysis	Date of Analysis
			14/06/18	26/07/18	05/10/18	23/01/19
			Initial	1st month	3rd month	6th month
1	Appearance	White or almost white powder	Almost white powder	Almost white powder	Almost white powder	Almost white powder

S. No.	Tests	Acceptance criteria	Date of Analysis	Date of Analysis	Date of Analysis	Date of Analysis
			14/06/18	26/07/18	05/10/18	23/01/19
			Initial	1st month	3rd month	6th month
2	Identification by IR	The IR spectrum of the sample should be concordant with the IR spectrum of Pantoprazole sodium sesquihydrate reference/ working standard.	Complies	Complies	Complies	Complies
3	Water by KF titrator (% w/w, use about 0.15 g of sample)	Between 5.9 and 6.9	6.50	6.43	6.40	6.42
4	Related substances by HPLC (% w/w) a) Impurity-A b) Impurity-B c) Impurity-C d) Impurity-(D+F) e) Impurity-E Unspecified impurity Total impurities	 Not more than 0.2 Not more than 0.1 Not more than 0.1 Not more than 0.2 Not more than 0.1 Not more than 0.10 Not more than 0.5	 0.06 Not detected Not detected Not detected 0.03 (BDL) 0.01 (BDL) 0.06	 0.08 BDL BDL BDL BDL BDL 0.08	 0.07 BDL BDL BDL BDL BDL 0.07	 0.08 BDL BDL 0.07 BDL BDL 0.16
5	Assay by Potentiometry(% w/w, on anhydrous basis)	*Not less than 99.0 and not more than 101.0*	99.7	99.6	99.8	100.4

Stability study report No.	X/10/06	**Reference protocol No.**	X/10	
Name of the Product	X	**Date of Mfg.**	Mar-18	
Batch No.	SL/18/X/002	**Date Kept for stability**	18/06/18	
Specification No.	X/EP/04	**STP/PTP No.**	X/EP/04	
Storage conditions	Pack the sample under inert atmosphere in transparent polyethylene, twist and tie, followed by black polyethylene bag, twist and tie and then place in a HDPE container.			
	Temperature: 40 ± 2 °C		**Relative Humidity : 75 ± 5%**	

S. No.	Tests	Acceptance criteria	Date of Analysis	Date of Analysis	Date of Analysis	Date of Analysis
			14/06/18	26/07/18	05/10/18	23/01/19
			Initial	1st month	3rd month	6th month
1	Appearance	White or almost white powder	Almost white powder	Almost white powder	Almost white powder	Almost white powder
2	Identification by IR	The IR spectrum of the sample should be concordant with the IR spectrum of Pantoprazole sodium sesquihydrate reference/ working standard.	Complies	Complies	Complies	Complies
3	Water by KF titrator (% w/w, use about 0.15 g of sample)	Between 5.9 and 6.9	6.46	6.60	6.42	6.28
4	Related substances by HPLC (% w/w) a) Impurity-A b) Impurity-B c) Impurity-C d) Impurity-(D+F) e) Impurity-E Unspecified impurity Total impurities	 Not more than 0.2 Not more than 0.1 Not more than 0.1 Not more than 0.2 Not more than 0.1 Not more than 0.10 Not more than 0.5	 0.09 Not detected Not detected Not detected 0.03 (BDL) 0.02 (BDL) 0.09	 0.11 BDL BDL BDL BDL BDL 0.11	 0.11 BDL BDL BDL BDL BDL 0.11	 0.12 BDL BDL 0.08 0.05 BDL 0.25
5	Assay by Potentiometry(% w/w, on anhydrous basis)	*Not less than 99.0 and not more than 101.0*	100.2	99.1	99.1	100.6

Stability study report No.	X/10/06	**Reference protocol No.**	X/10	
Name of the Product	X	**Date of Mfg.**	Mar-18	
Batch No.	X/S00074/074	**Date Kept for stability**	18/06/18	
Specification No.	X/EP/04	**STP/PTP No.**	X/EP/04	
Storage conditions	Pack the sample under inert atmosphere in transparent polyethylene, twist and tie, followed by black polyethylene bag, twist and tie and then place in a HDPE container.			
	Temperature: 40 ± 2 °C		**Relative Humidity : 75 ± 5%**	

S. No.	Tests	Acceptance criteria	Date of Analysis	Date of Analysis	Date of Analysis	Date of Analysis
			14/06/18	26/07/18	05/10/18	23/01/19
			Initial	1st month	3rd month	6th month
1	Appearance	White or almost white powder	Almost white powder	Almost white powder	Almost white powder	Almost white powder
2	Identification by IR	The IR spectrum of the sample should be concordant with the IR spectrum of Pantoprazole sodium sesquihydrate reference/ working standard.	Complies	Complies	Complies	Complies
3	Water by KF titrator (% w/w, use about 0.15 g of sample)	Between 5.9 and 6.9	6.48	6.28	6.31	6.38
4	Related substances by HPLC (% w/w) a) Impurity-A b) Impurity-B c) Impurity-C d) Impurity-(D+F) e) Impurity-E Unspecified impurity Total impurities	 Not more than 0.2 Not more than 0.1 Not more than 0.1 Not more than 0.2 Not more than 0.1 Not more than 0.10 Not more than 0.5	 0.06 Not detected Not detected Not detected 0.04 (BDL) 0.02 (BDL) 0.06	 0.06 BDL BDL BDL BDL BDL 0.06	 0.07 BDL BDL BDL BDL BDL 0.07	 0.07 BDL BDL 0.07 BDL 0.05 0.18
5	Assay by Potentiometry(% w/w, on anhydrous basis)	*Not less than 99.0 and not more than 101.0*	100.8	99.9	99.9	100.1

5. Conclusion

As per above discussion the six months stability study performed on product X drug and its results are summarized. It clearly shows that all parameter found within predefined limit. Hence we can say that the product is stable in the proposed container closer system at long term condition up to six months.

References:

1) International Conference on Harmonization: Guidelines on stability testing of new drug substances and Products (Q1A (R2)), Feb 2013.

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3) HPLC instrumentation taken from http://en.wikipedia.org/wiki/Highperformance_ liquid chromatography accessed on 1/04/2014.

4) International Conference on Harmonization (1999) Specifications, Q6A: Test Procedures and Acceptance Criteria for New Drug Substances and New Drug Products. Chemical substances 65 (146):67488.

5) G.H Jefferey, J Bassett, et.al, "Vogel's Textbook of Quantitative Chemical analysis", Fifth edition, 1989

6) "Instrumental chemical analysis: Basic principles and techniques", Department of chemical and bimolecular engineering, National university of Singapore